

Exact POMDP Solutions: α -vectors

Recap

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- POMDP

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- POMDP

$$(S, A, O, R, T, Z, \gamma)$$

Recap

- POMDP $(S, A, O, R, T, Z, \gamma)$
- Belief Updates

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$$b_t(s) = P(s_t = s \mid h_t)$$

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- POMDP $(S, A, O, R, T, Z, \gamma)$
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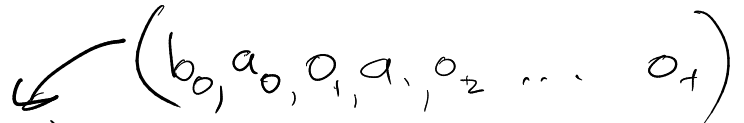
$$b_t(s) = P(s_t = s \mid h_t)$$

$$b' = \tau(b, a, o)$$

Recap

- POMDP $(S, A, O, R, T, Z, \gamma)$
- Belief Updates

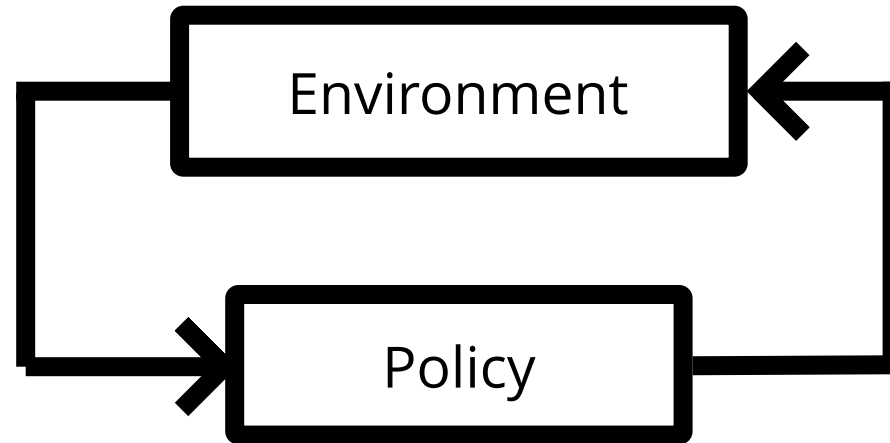
$$b_t(s) = P(s_t = s \mid h_t)$$

 $(b_0, a_0, o_1, a_1, o_2, \dots, o_t)$

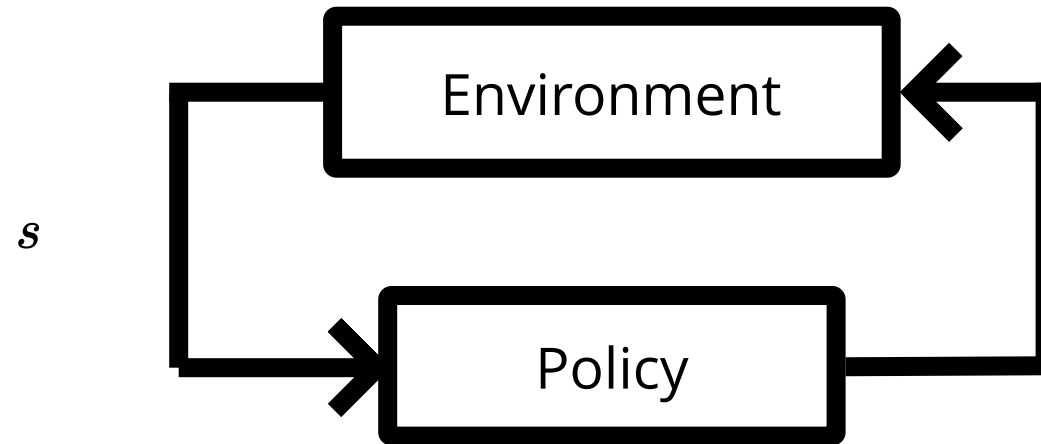
$$b' = \tau(b, a, o)$$

$$b'(s') \propto Z(o \mid a, s') \sum_s T(s' \mid s, a) b(s)$$

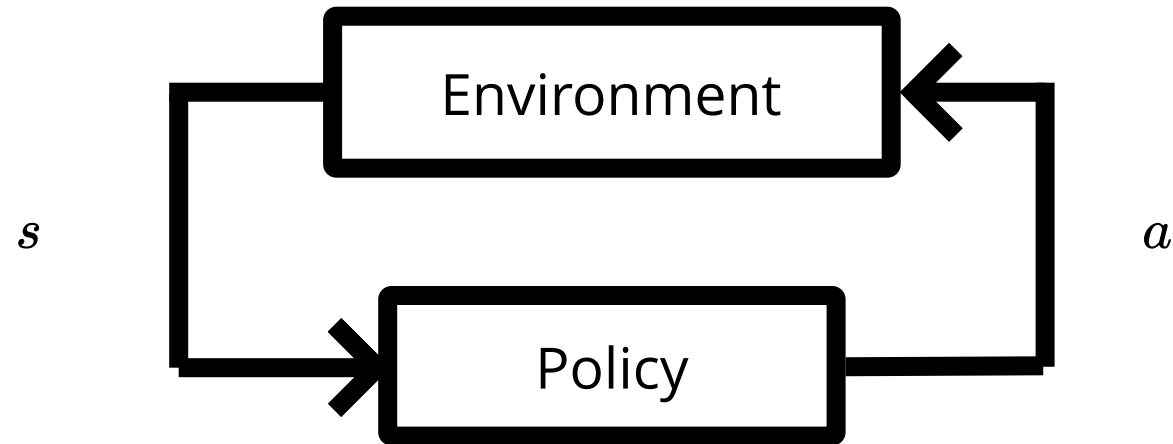
MDP Sense-Plan-Act Loop



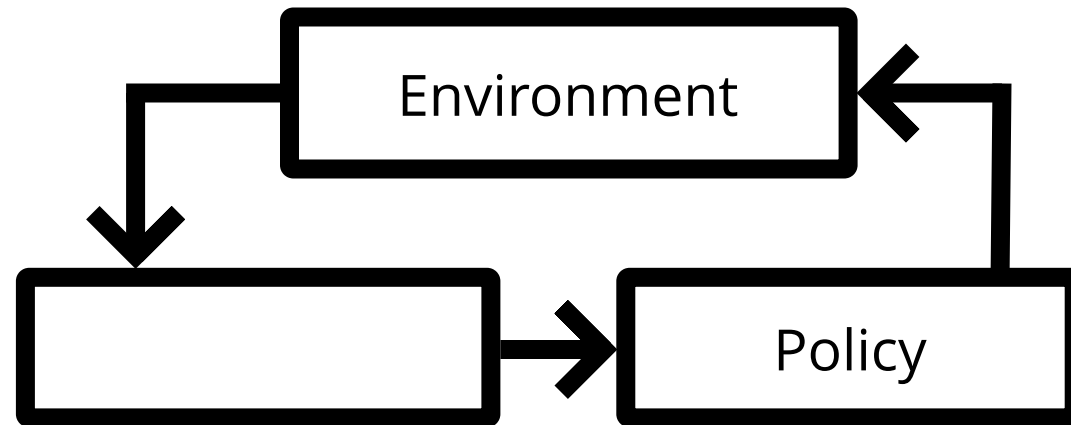
MDP Sense-Plan-Act Loop



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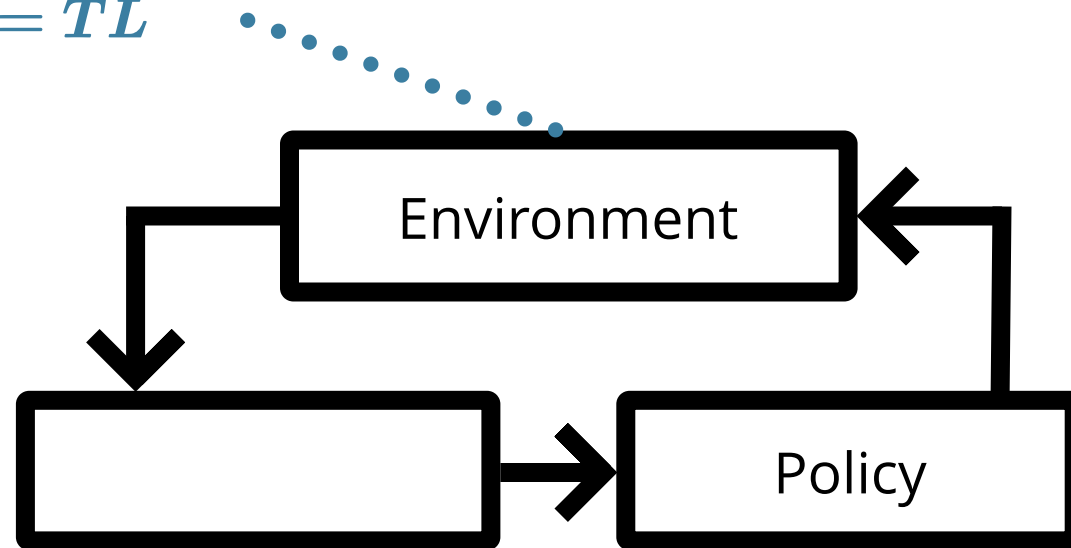
POMDP Sense-Plan-Act Loop



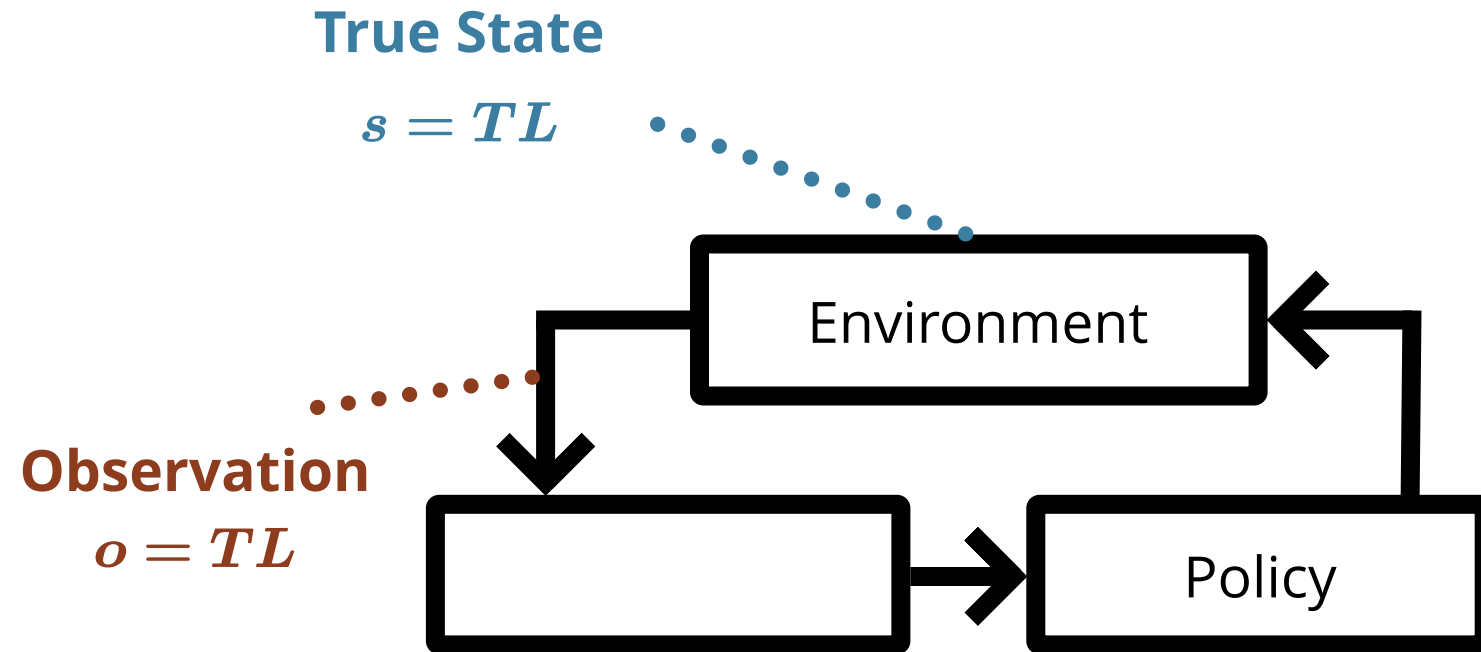
POMDP Sense-Plan-Act Loop

True State

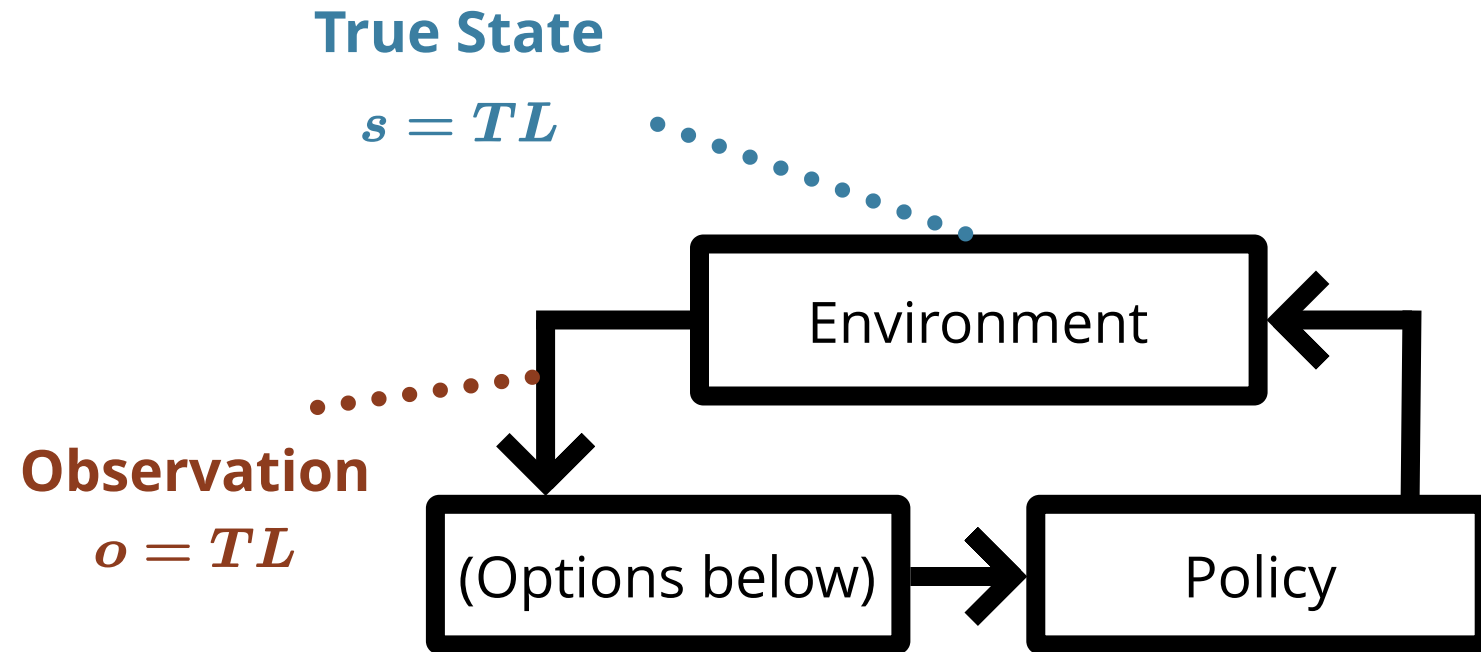
$$s = TL$$



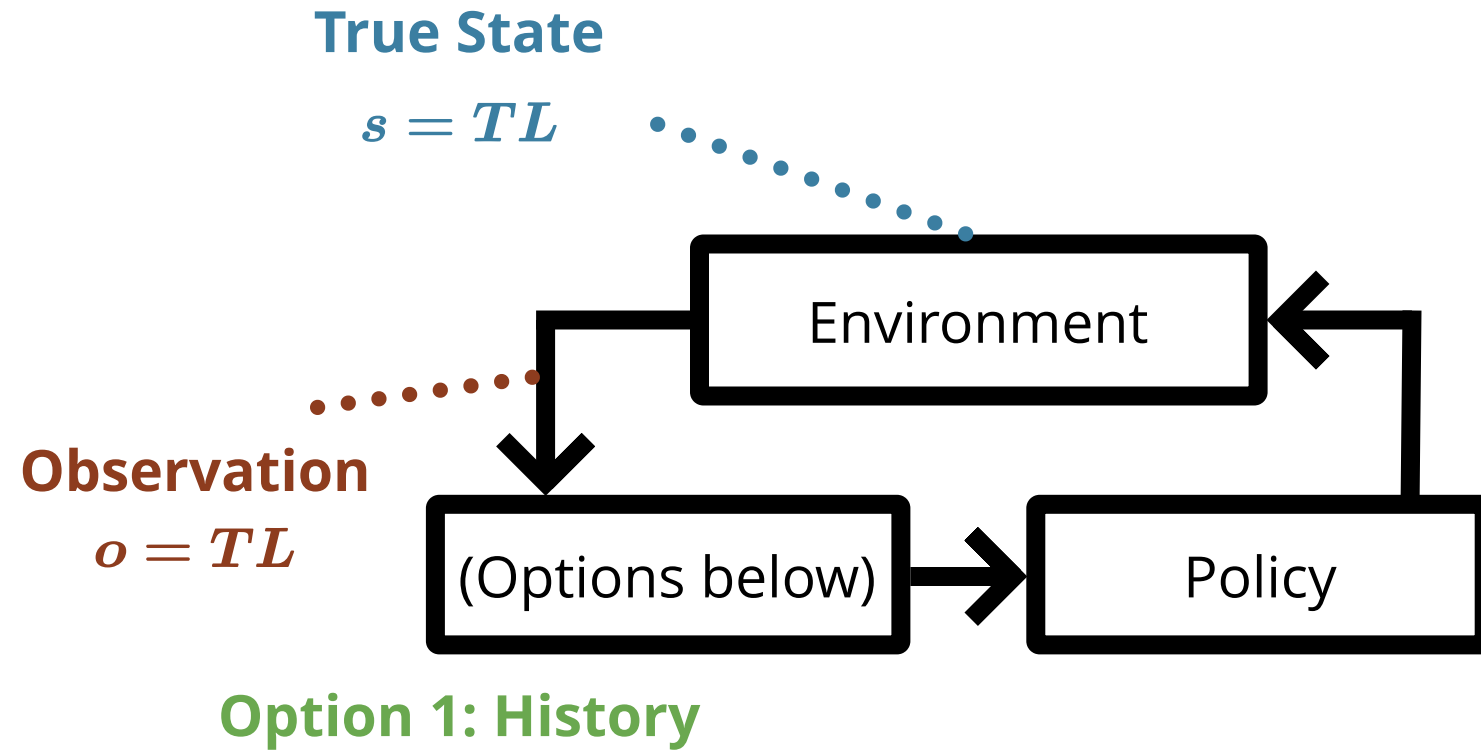
POMDP Sense-Plan-Act Loop



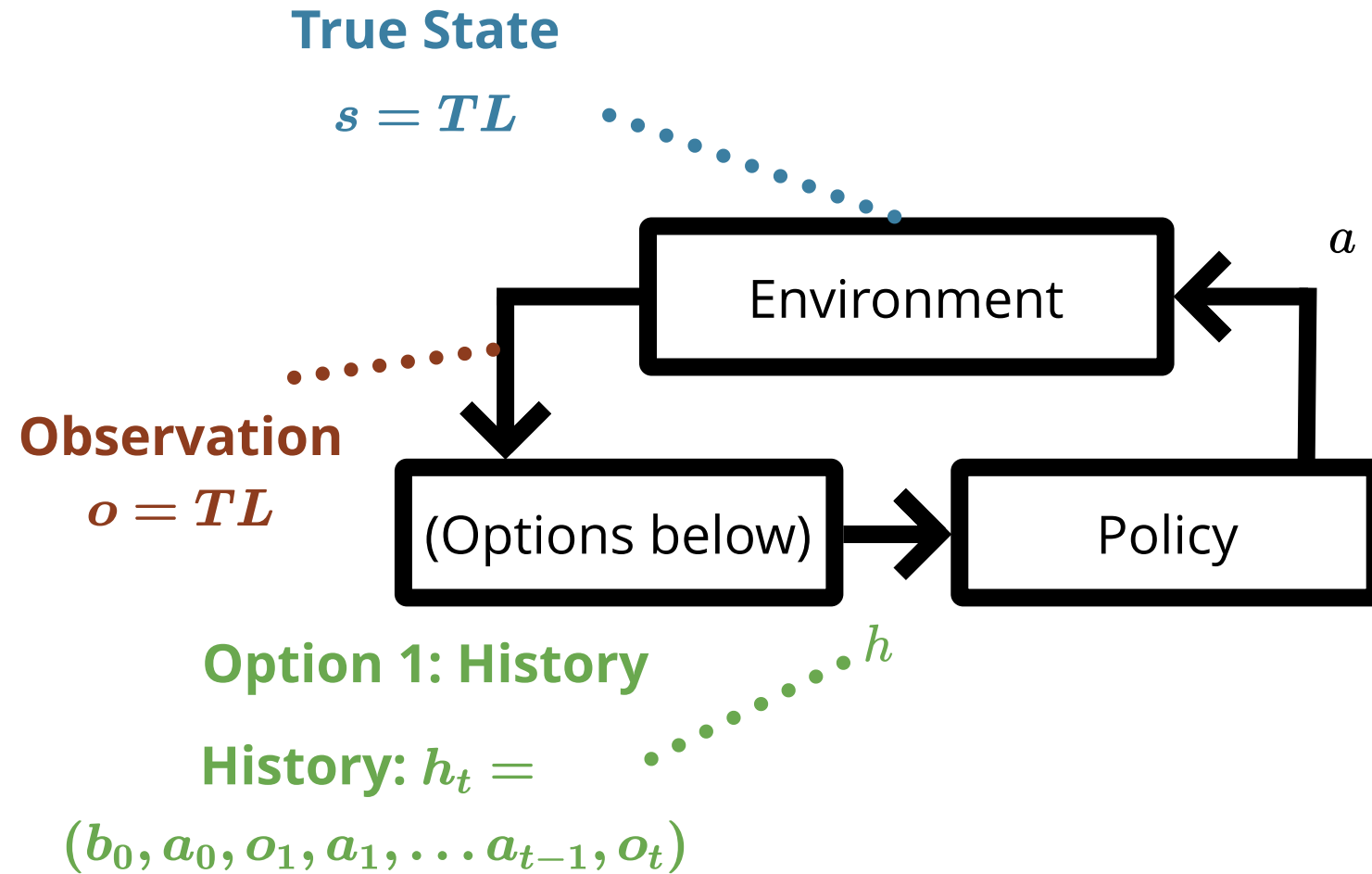
POMDP Sense-Plan-Act Loop



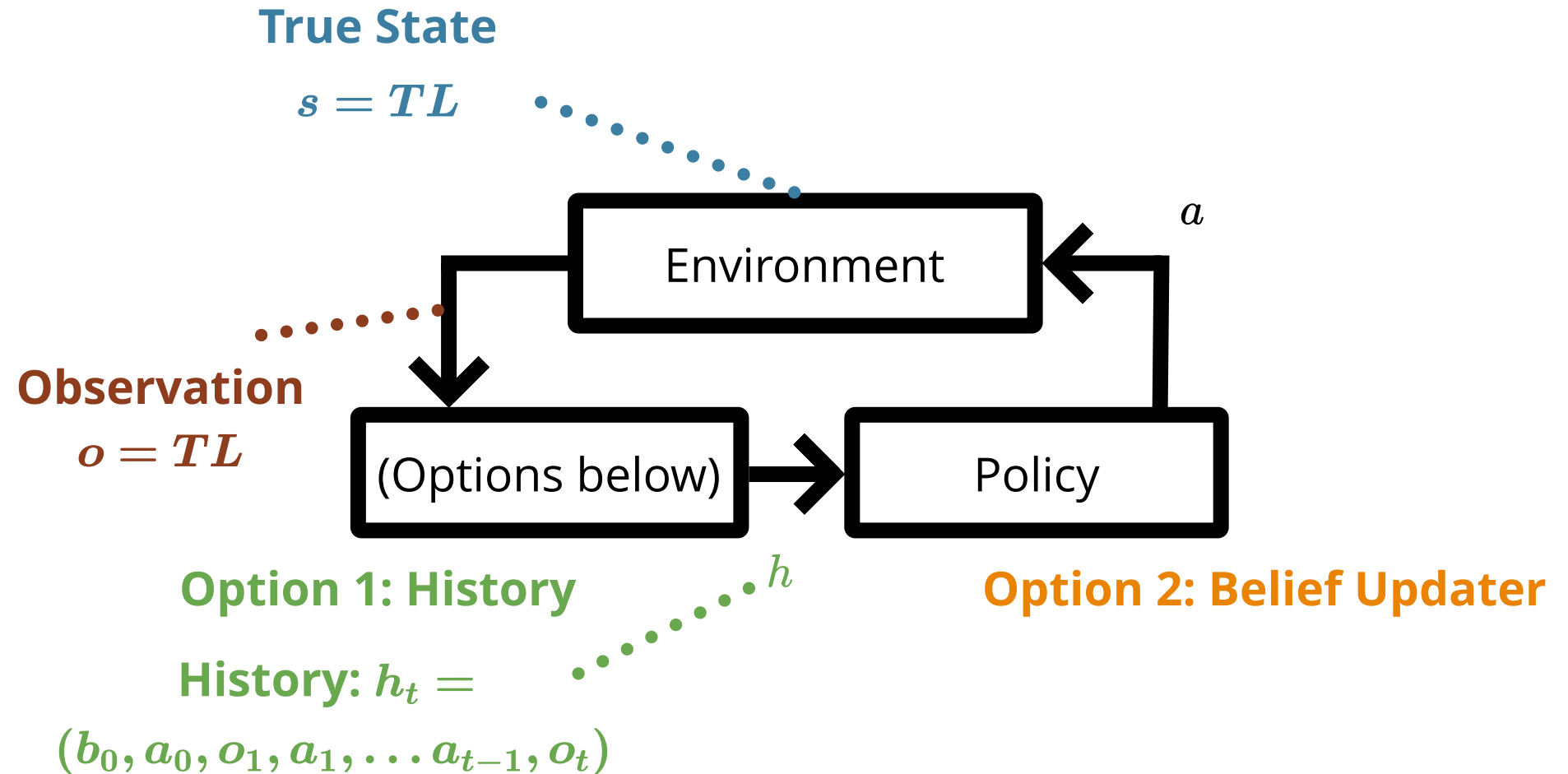
POMDP Sense-Plan-Act Loop



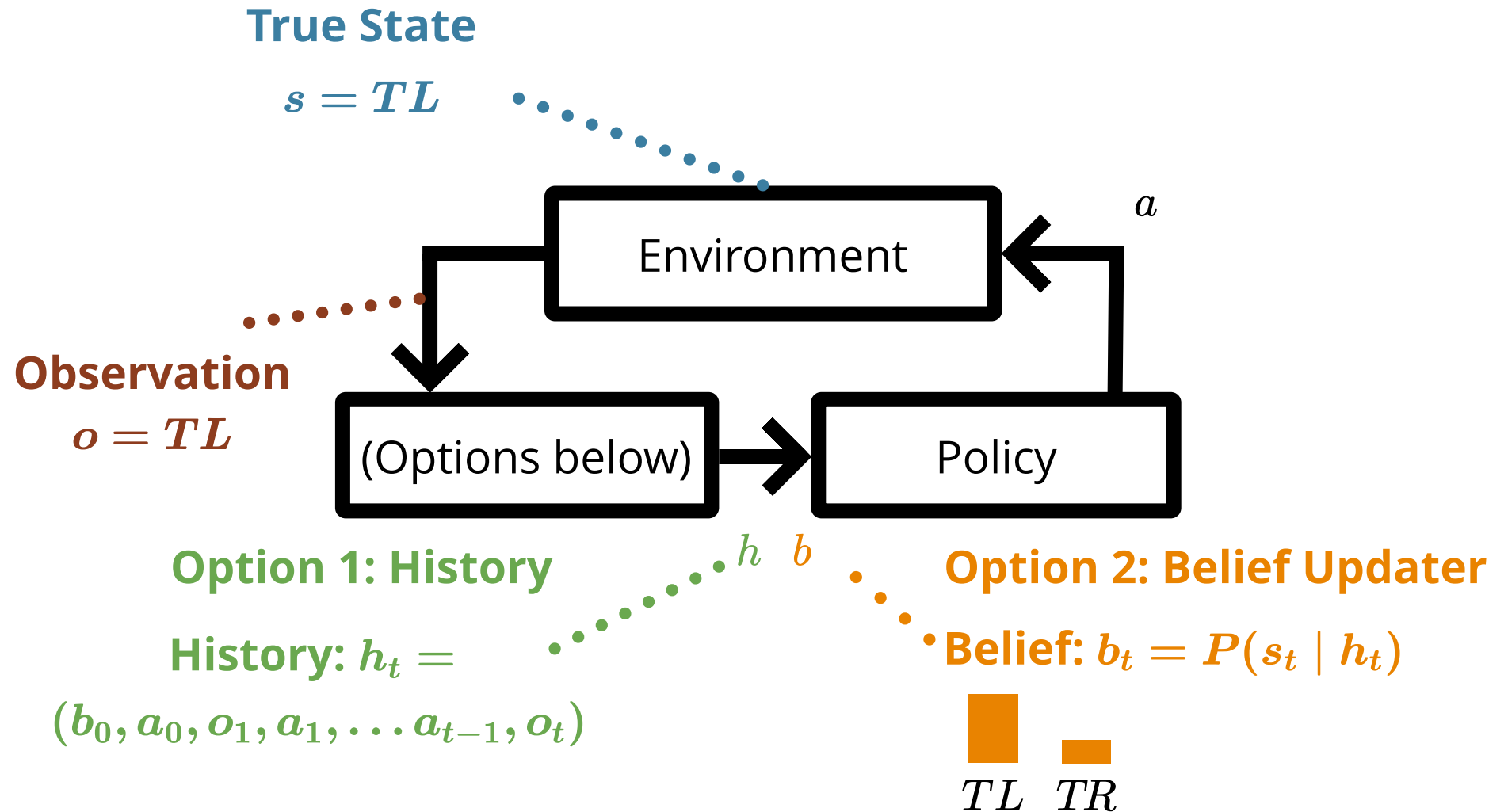
POMDP Sense-Plan-Act Loop



POMDP Sense-Plan-Act Loop



POMDP Sense-Plan-Act Loop



Exercise 1: Crying Baby Belief Update

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$$S = \{h, \overset{\text{not}}{\neg} h\}$$

$$A = \{f, \neg f\}$$

$$O = \{c, \neg c\}$$

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$$R(s, a) = R(s) + R(a)$$

Exercise 1: Crying Baby Belief Update

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$$R(s) = \begin{cases} -10 & \text{if } s = h \\ 0 & \text{otherwise} \end{cases}$$

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Exercise 1: Crying Baby Belief Update

$$S = \{h, \neg h\} \quad T(h \mid h, \neg f) = 1.0$$

$$A = \{f, \neg f\} \quad T(h \mid \neg h, \neg f) = 0.1$$

$$O = \{c, \neg c\} \quad T(\neg h \mid \cdot, f) = 1.0$$

$$R(s, a) = R(s) + R(a)$$

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$$b'(s') \propto Z(o \mid a, s') \sum_s T(s' \mid s, a) b(s)$$

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$$Z(c \mid \cdot, h) = 0.8$$

$$Z(c \mid \cdot, \neg h) = 0.1$$

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$$\gamma = 0.9$$

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$$A = \{f, \neg f\}$$

$$T(h \mid \neg h, \neg f) = 0.1$$

$$O = \{c, \neg c\}$$

$$T(\neg h \mid \cdot, f) = 1.0$$

$$b'(s') \propto \underbrace{Z(o \mid a, s')} \sum_s T(s' \mid s, a) b(s)$$

Starting at a $b(h) = 0$, calculate

b' with $a = \neg f$ and $o = c$.

$$R(s, a) = R(s) + R(a)$$

$$R(s) = \begin{cases} -10 & \text{if } s = h \\ 0 & \text{otherwise} \end{cases}$$

$$R(a) = \begin{cases} -5 & \text{if } a = f \\ 0 & \text{otherwise} \end{cases}$$

$$Z(c \mid \cdot, h) = 0.8$$

$$Z(c \mid \cdot, \neg h) = 0.1$$

$$\gamma = 0.9$$

$$b'(h) \propto Z(c \mid \neg f, h) (T(h \mid h, \neg f) b(h) + T(h \mid \neg h, \neg f) b(\neg h))$$

$$0.8 (1.0 \cdot 0.0 + 0.1 \cdot 1.0)$$

$$\rightarrow b'(h) \propto 0.08$$

$$b'(\neg h) \propto Z(c \mid \neg f, \neg h) (T(\neg h \mid h, \neg f) b(h) + T(\neg h \mid \neg h, \neg f) b(\neg h))$$

$$0.1 (0.0 + 1.0)$$

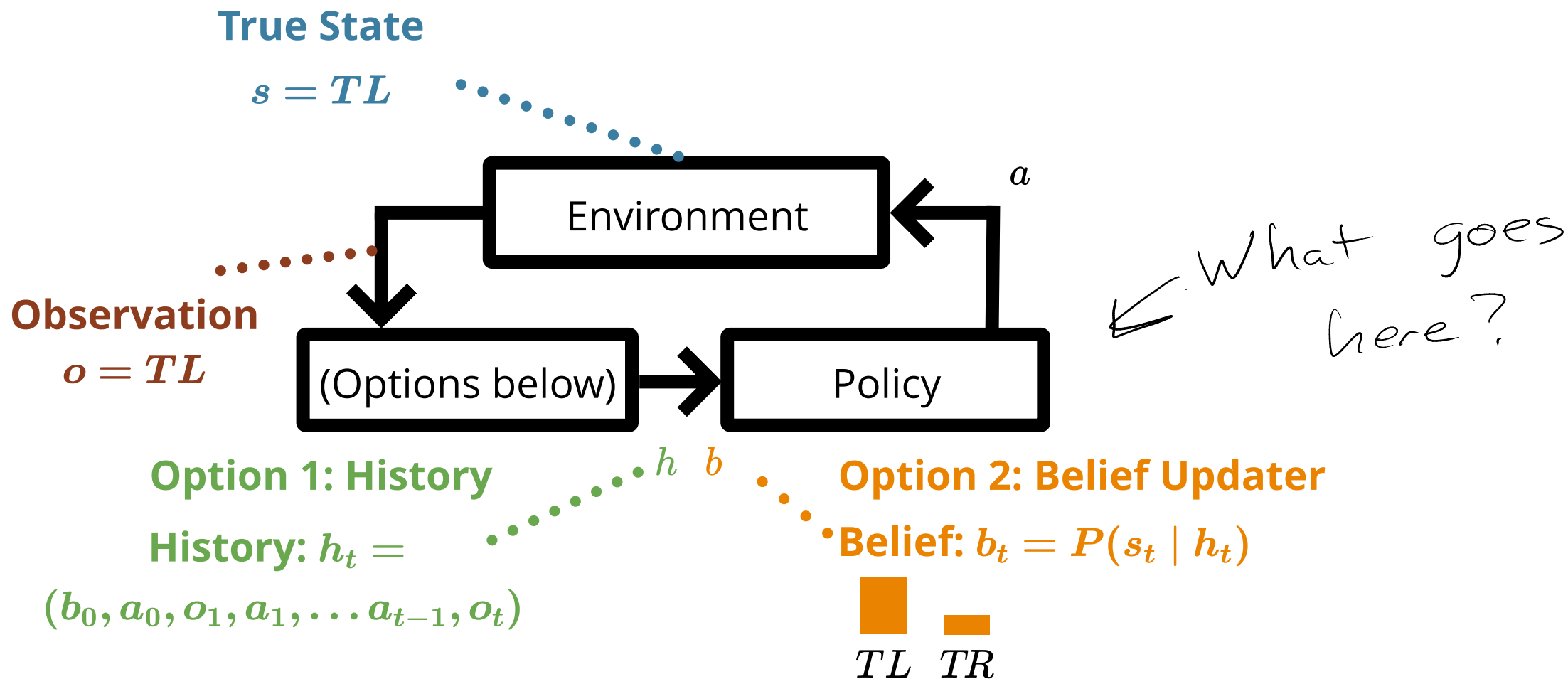
$$\rightarrow b'(\neg h) \propto 0.09$$

$$b'(h) = \frac{0.08}{0.08 + 0.09} = 47\%$$

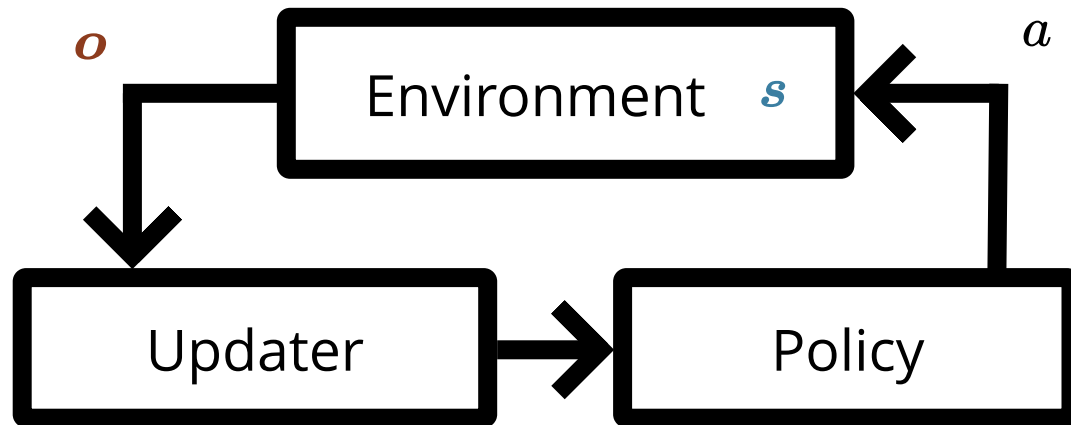
$$b'(\neg h) = 1 - 47\% = 53\% = \frac{0.09}{0.08 + 0.09}$$

Guiding Question

How do we calculate the optimal action in a POMDP?



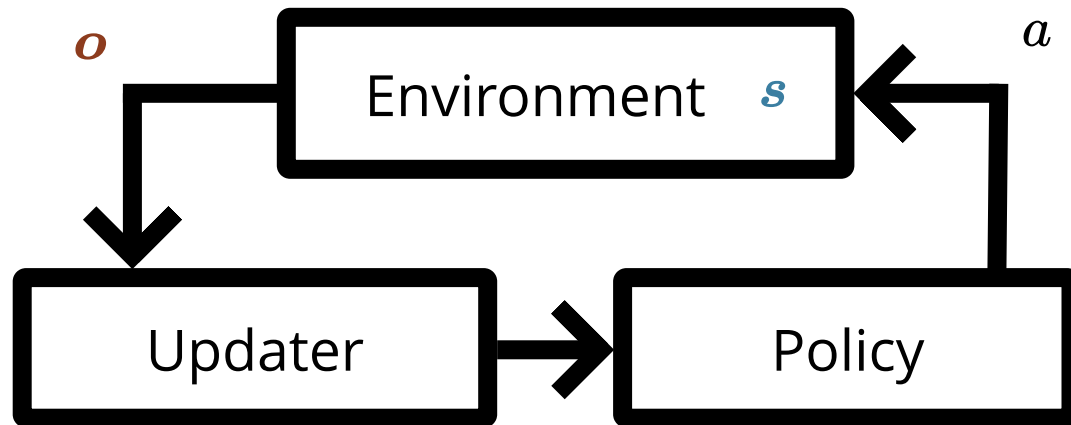
Belief-Space MDP



$$b_t = P(s_t | h_t)$$

 TL  TR

Belief-Space MDP

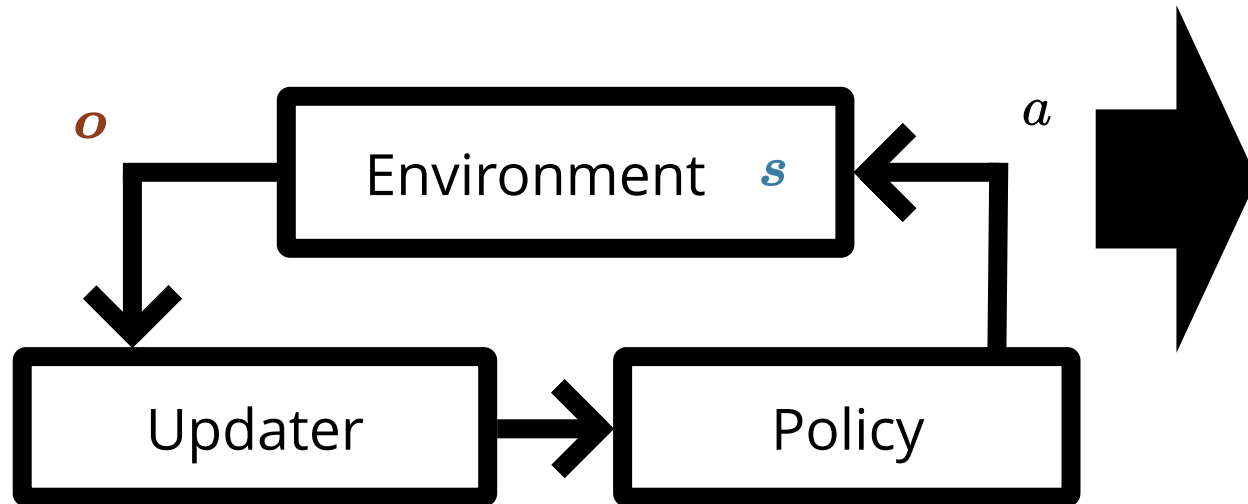


$$b_t = P(s_t | h_t)$$


TL


TR

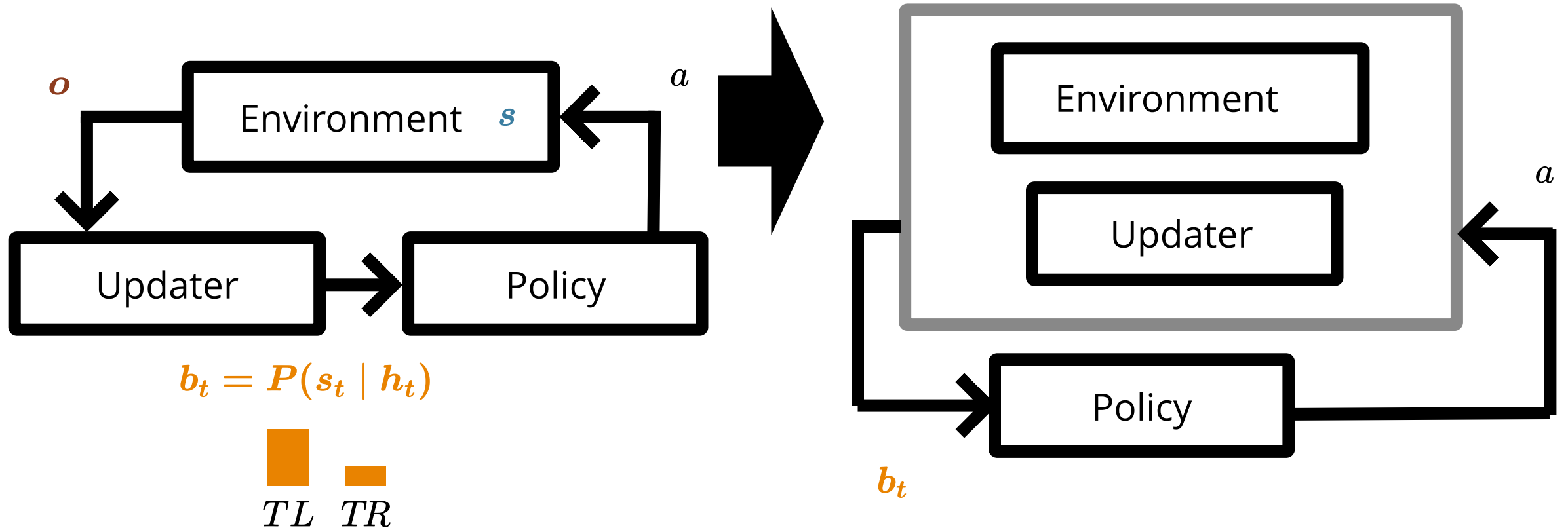
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Belief Dynamics

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$$b'(s') \propto Z(o \mid a, s') \sum_s T(s' \mid s, a) b(s)$$

Belief Dynamics

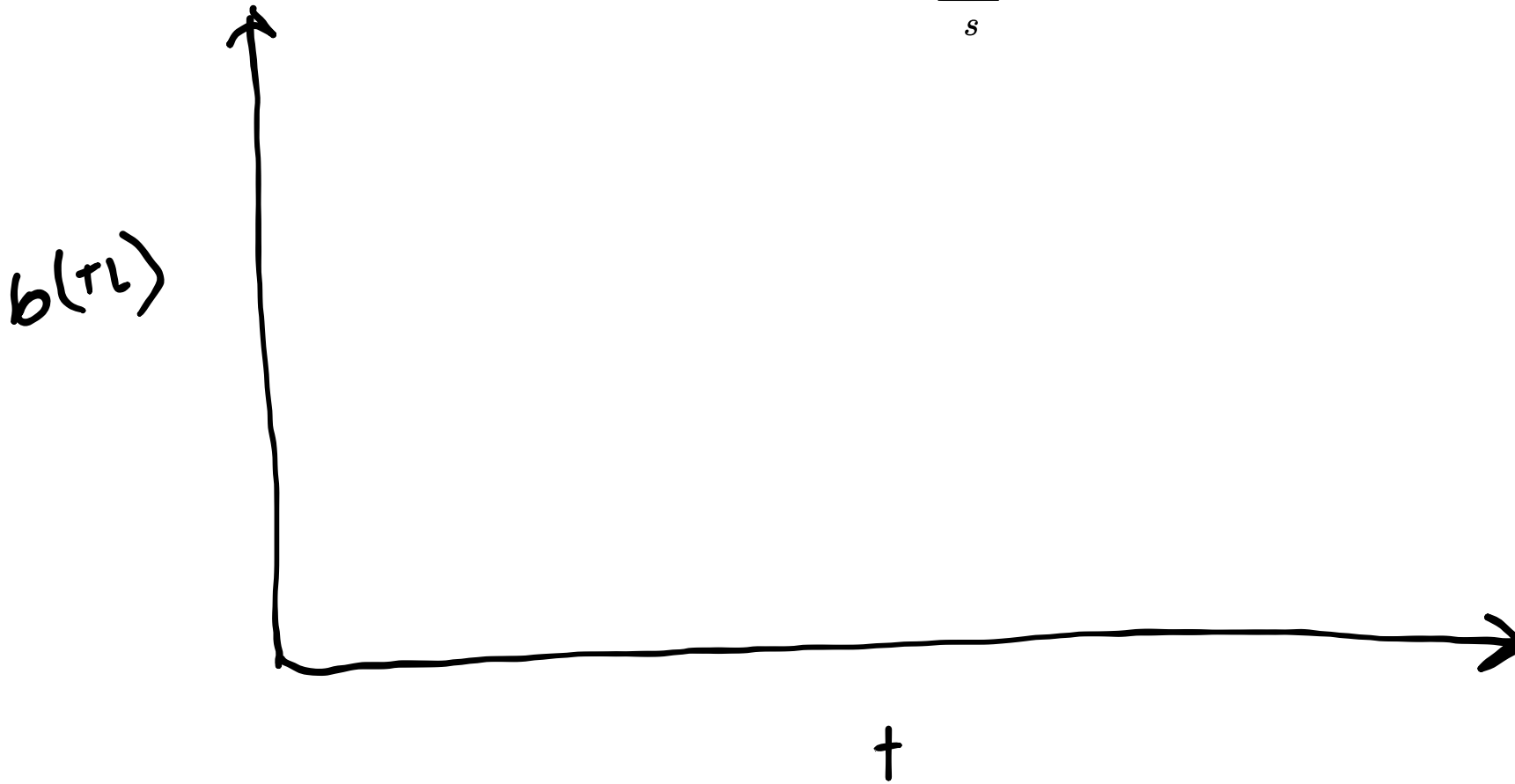
$$b'(s') \propto Z(o \mid a, s') \sum_s T(s' \mid s, a) b(s)$$

$b(\tau_L)$

†

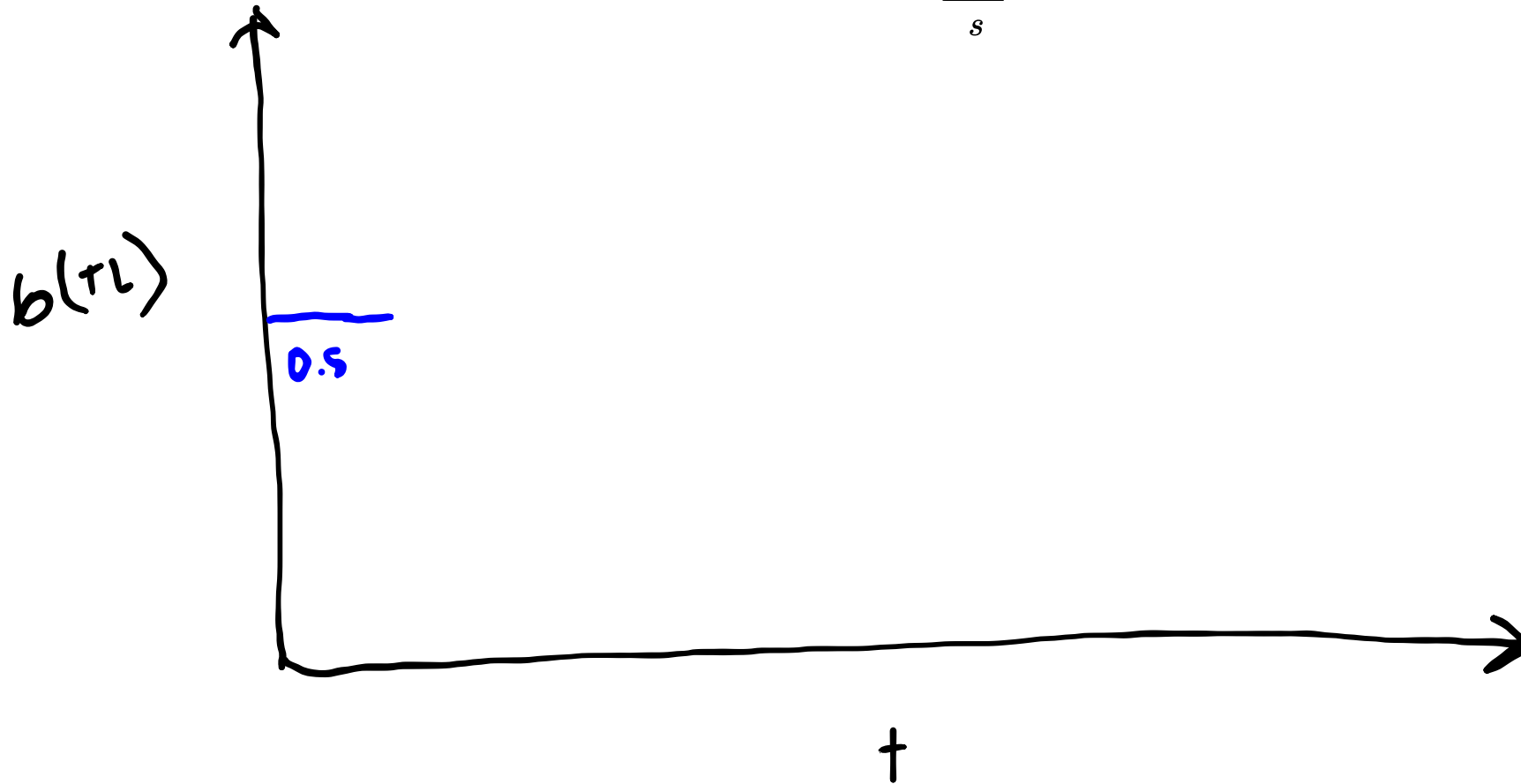
Belief Dynamics

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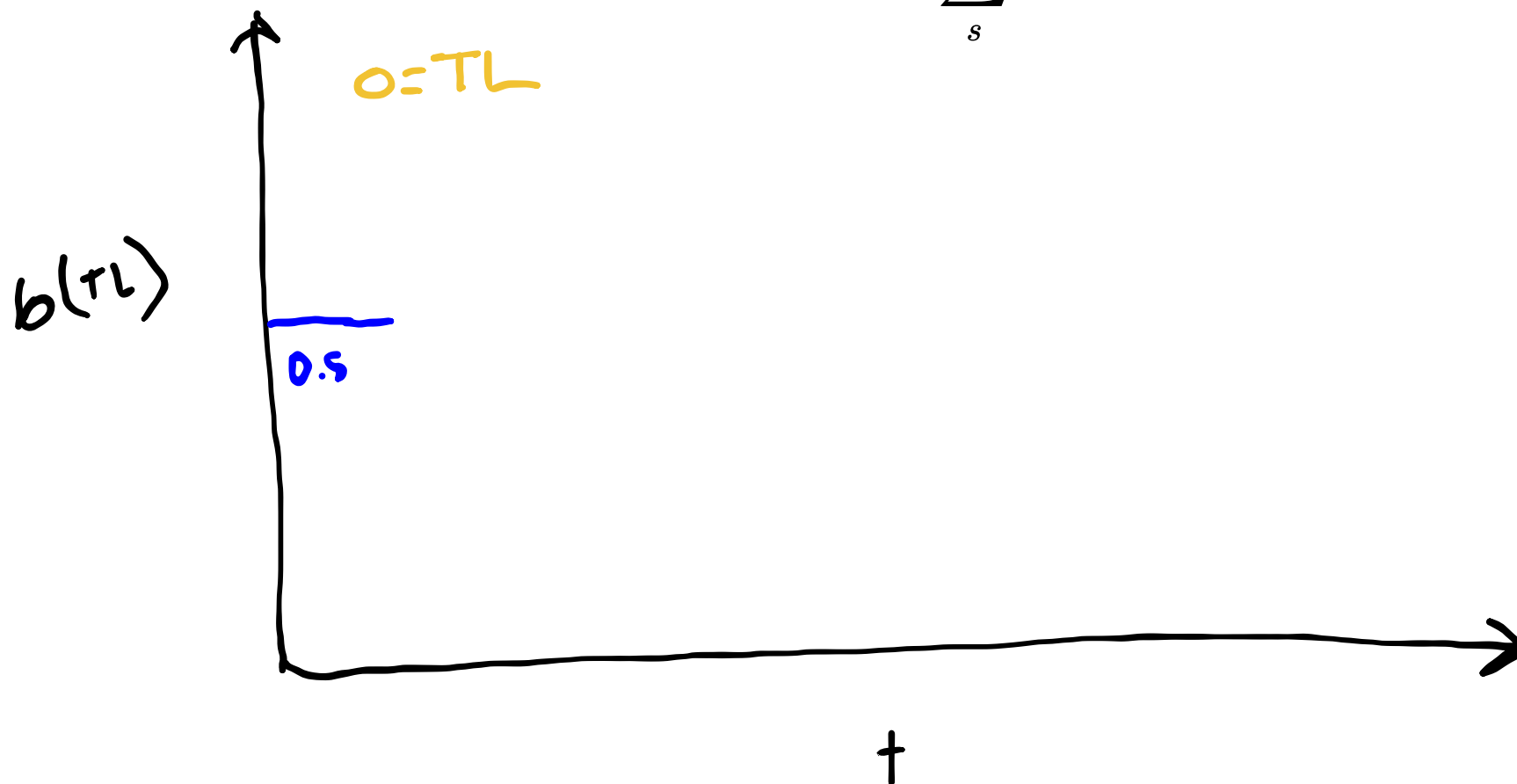
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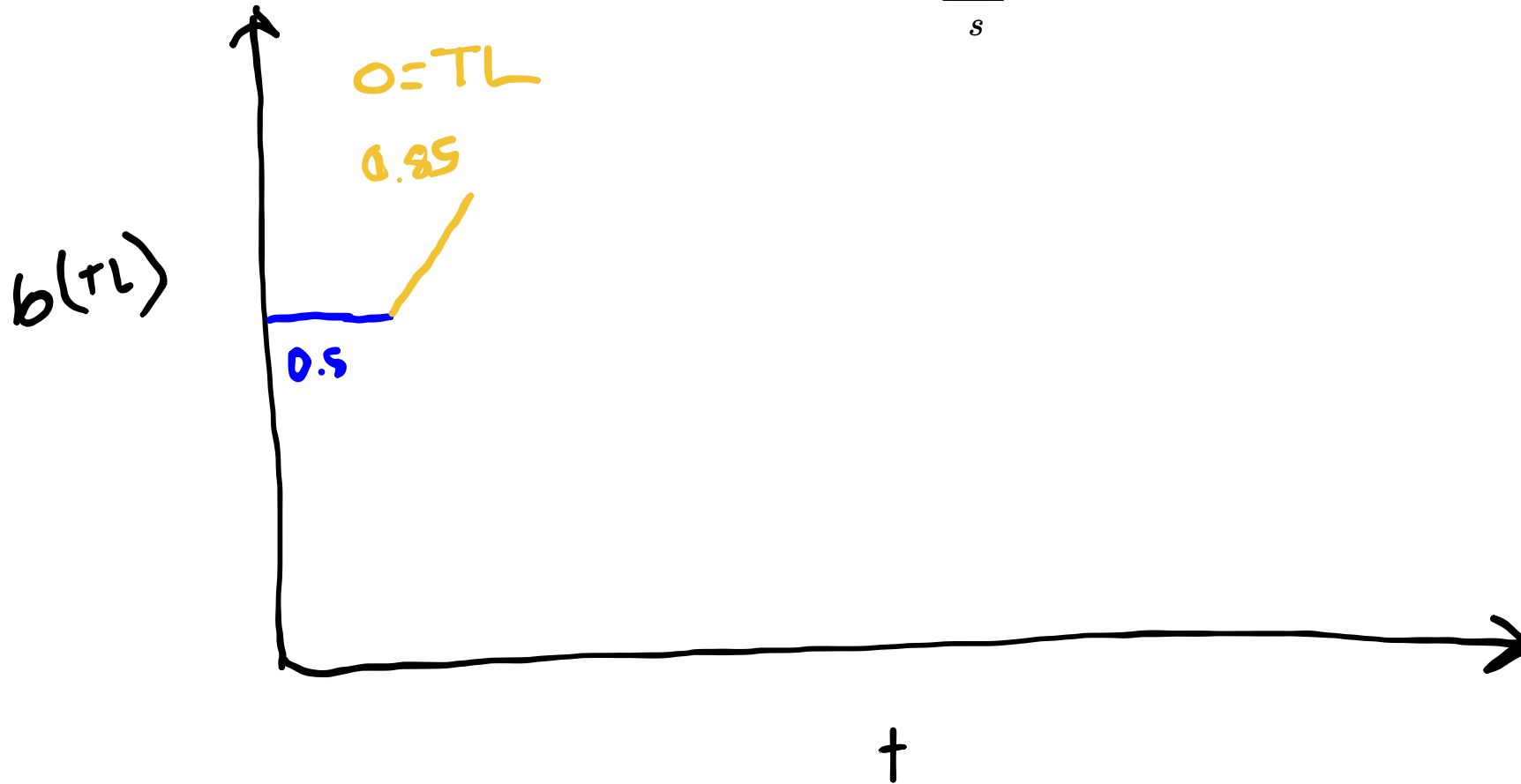
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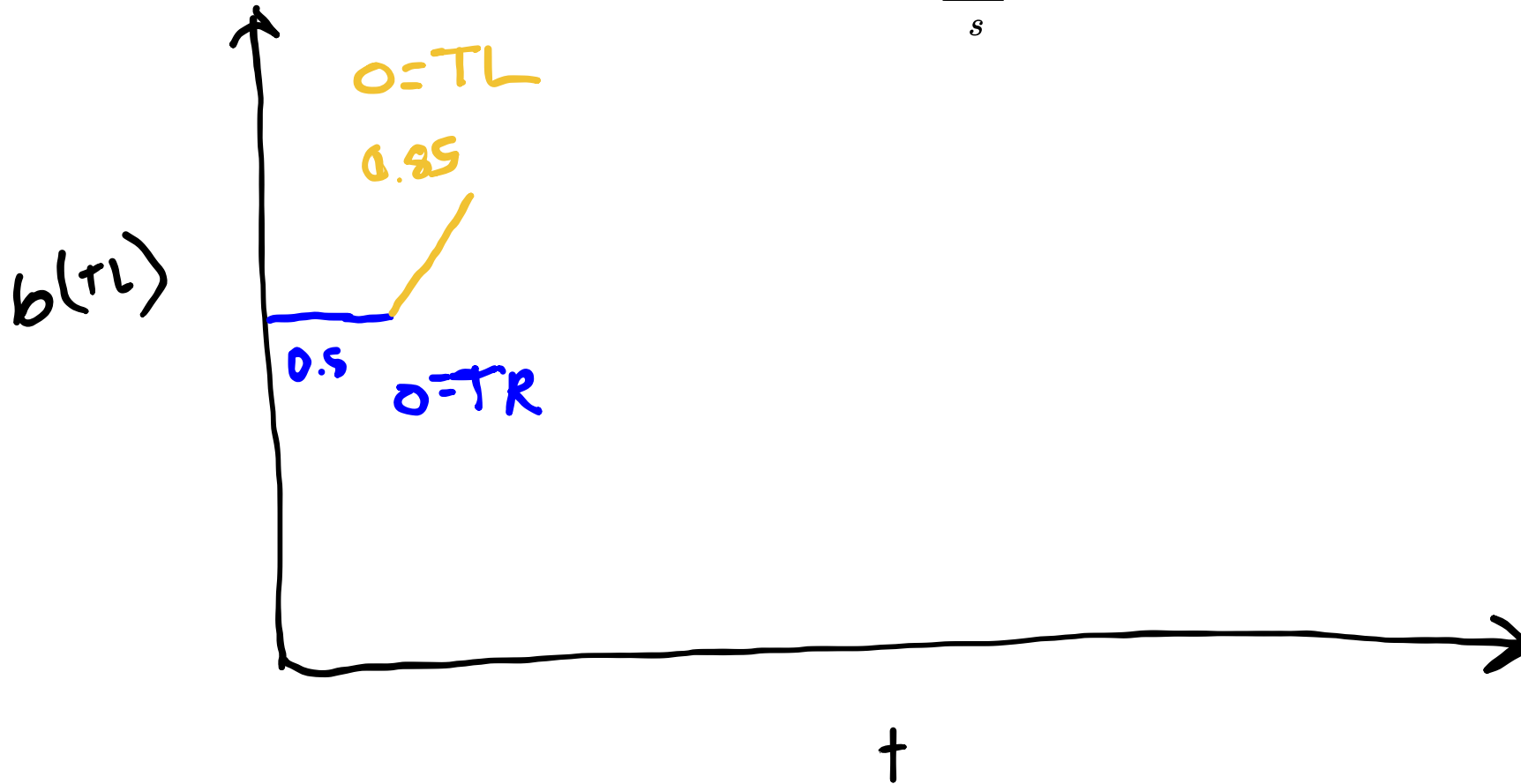
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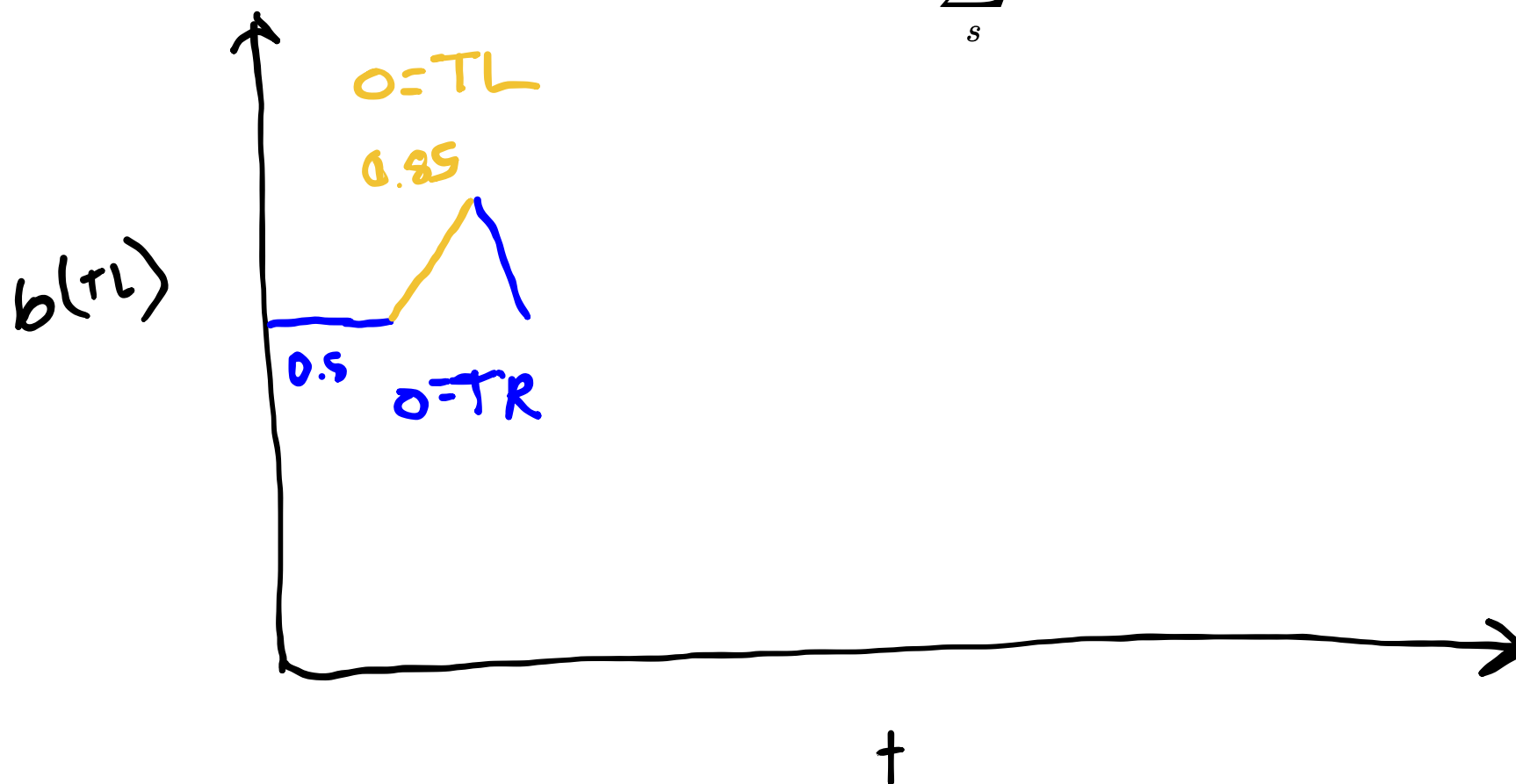
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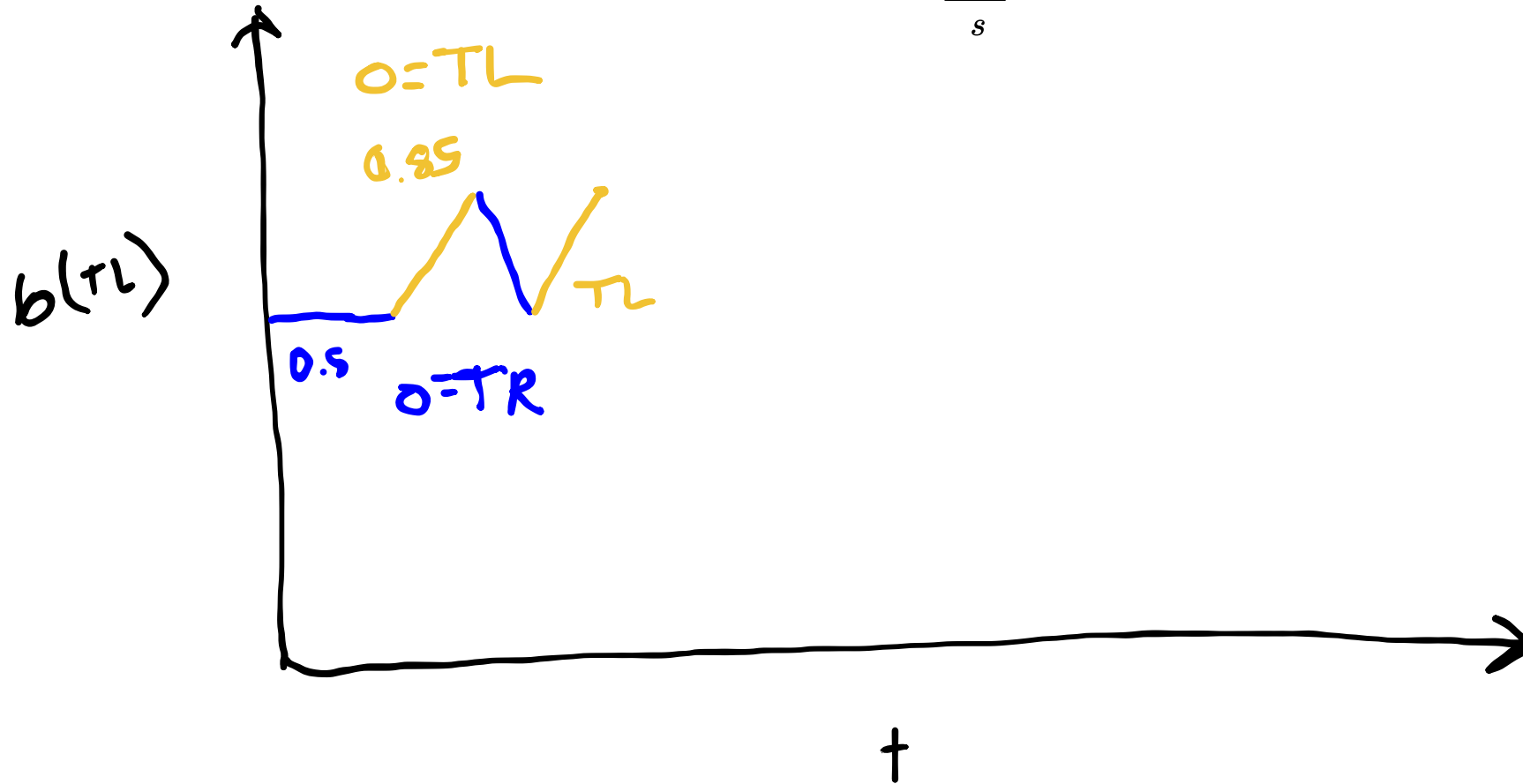
Belief Dynamics

$$b'(s') \propto Z(o | a, s') \sum_s T(s' | s, a) b(s)$$



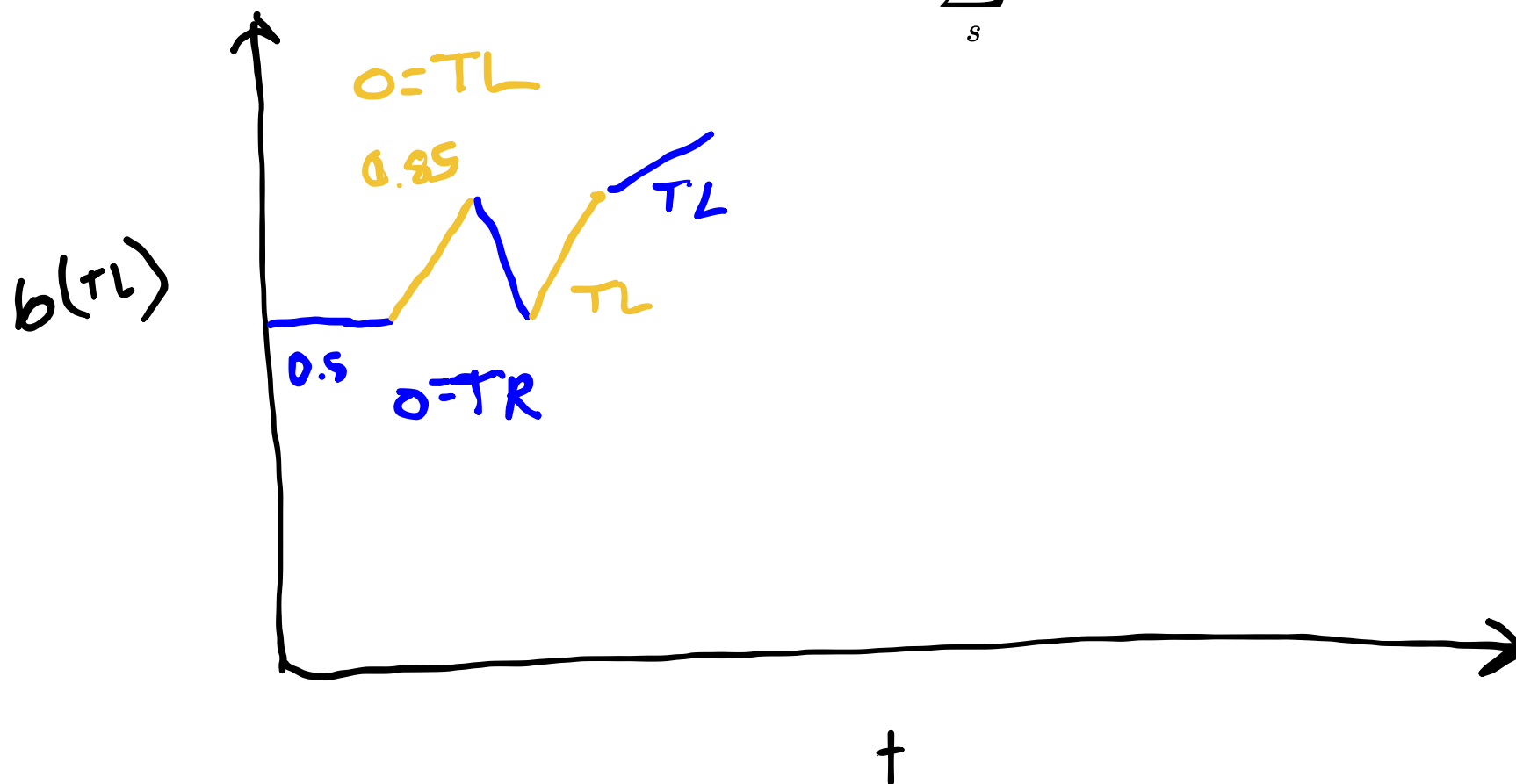
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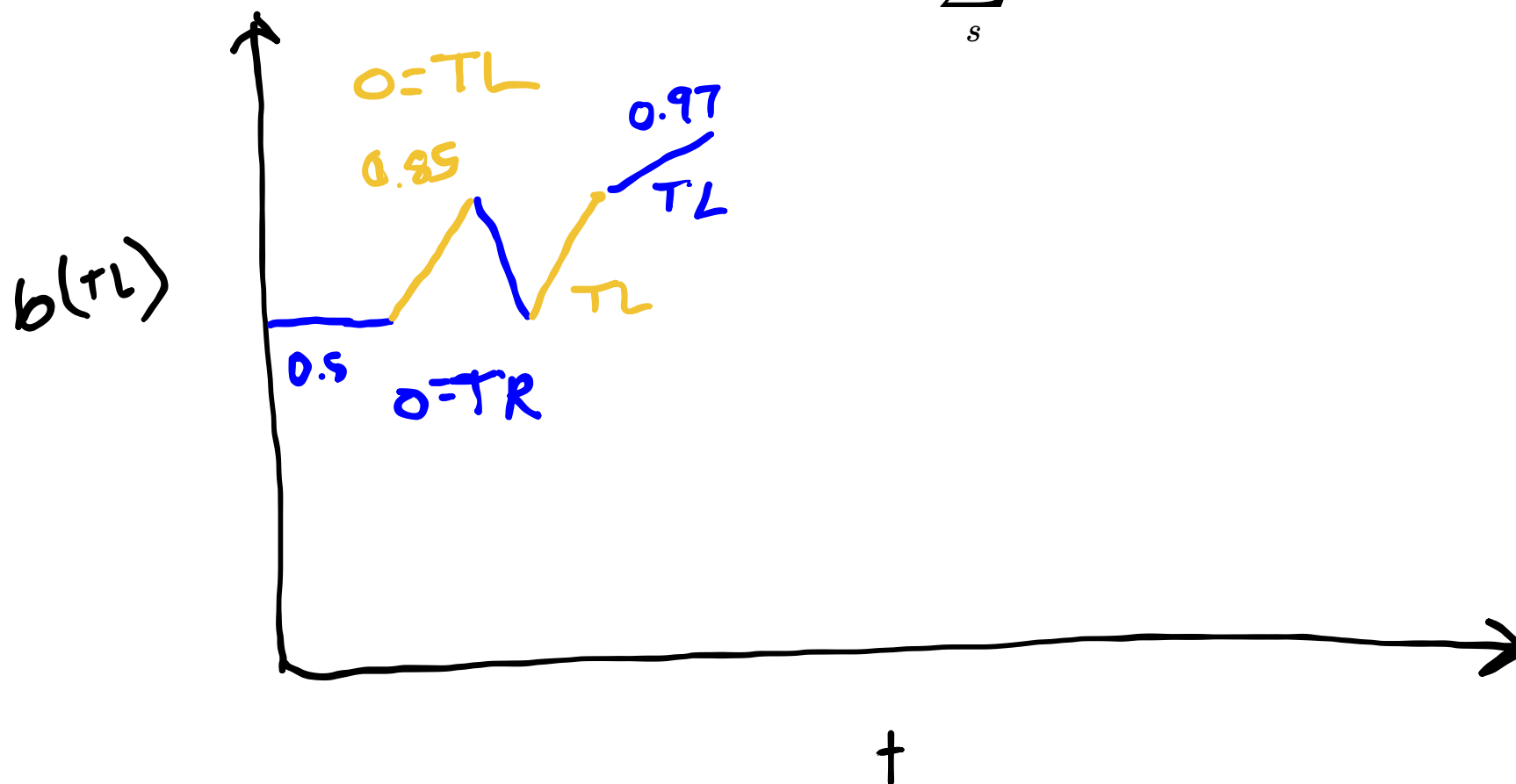
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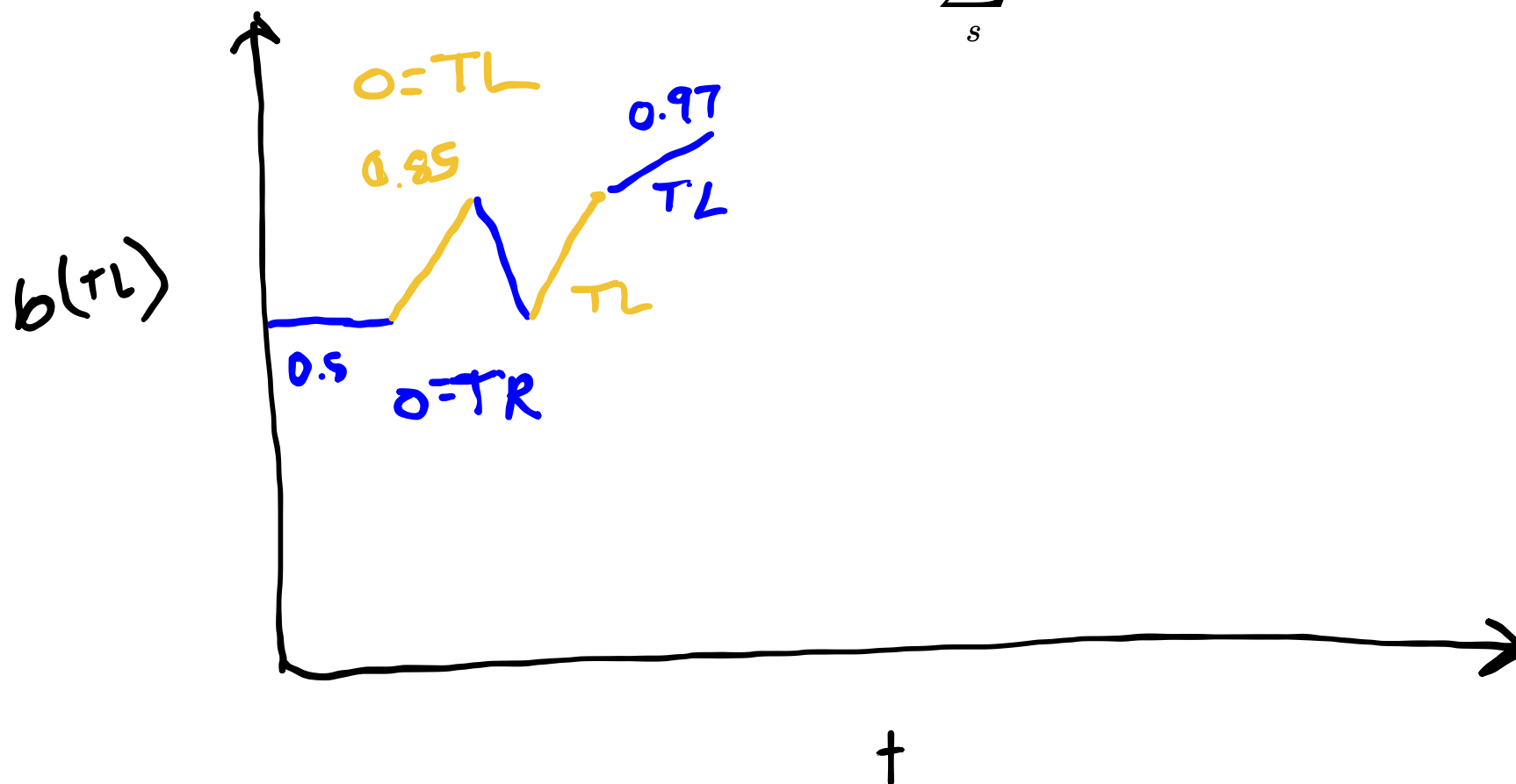
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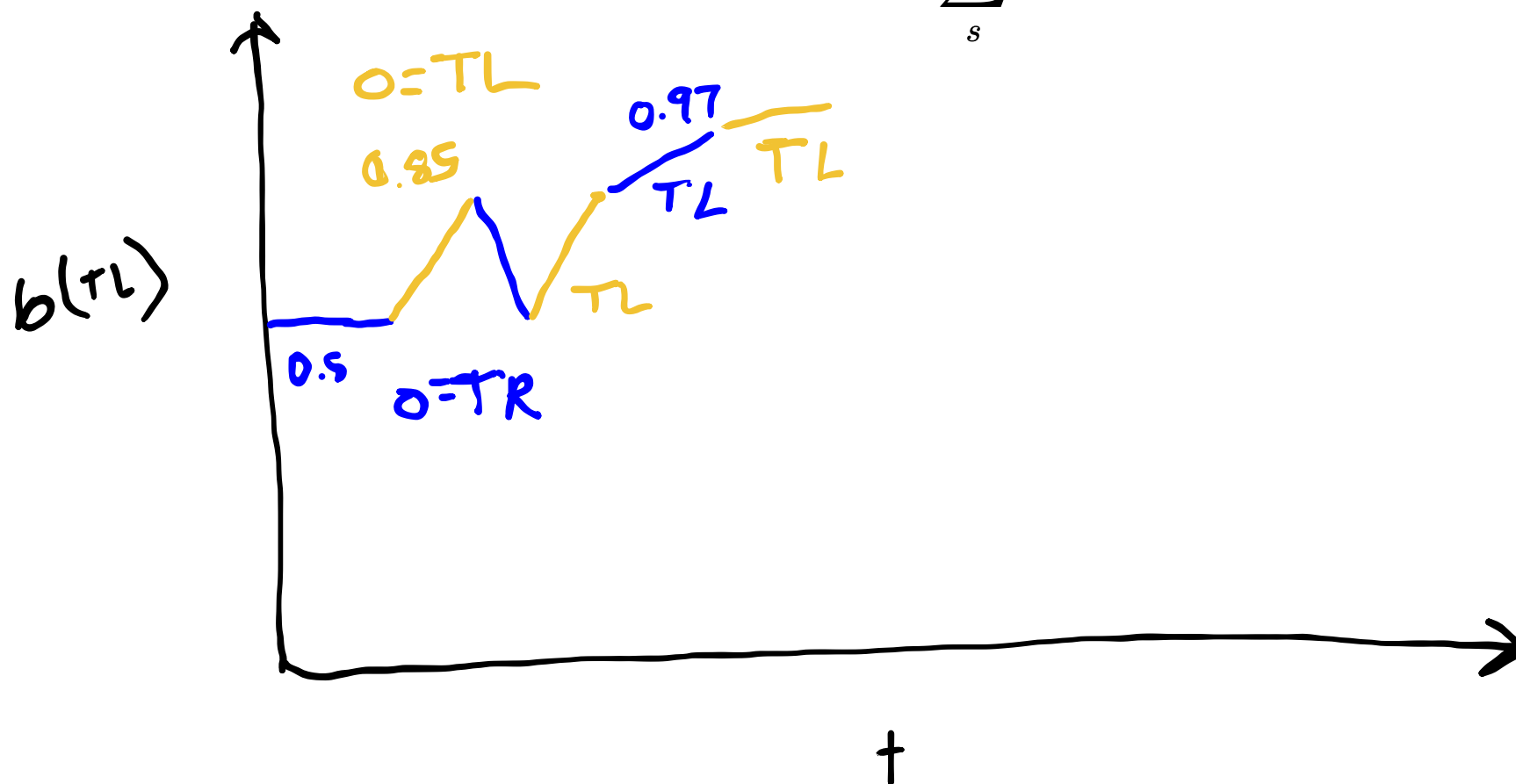
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Belief Dynamics

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Reward

- +10 opening empty
- 1 listening ←
- 100 opening tiger door
- 0 do nothing

One-step utility

(reward)

Want
 $R(b, a)$

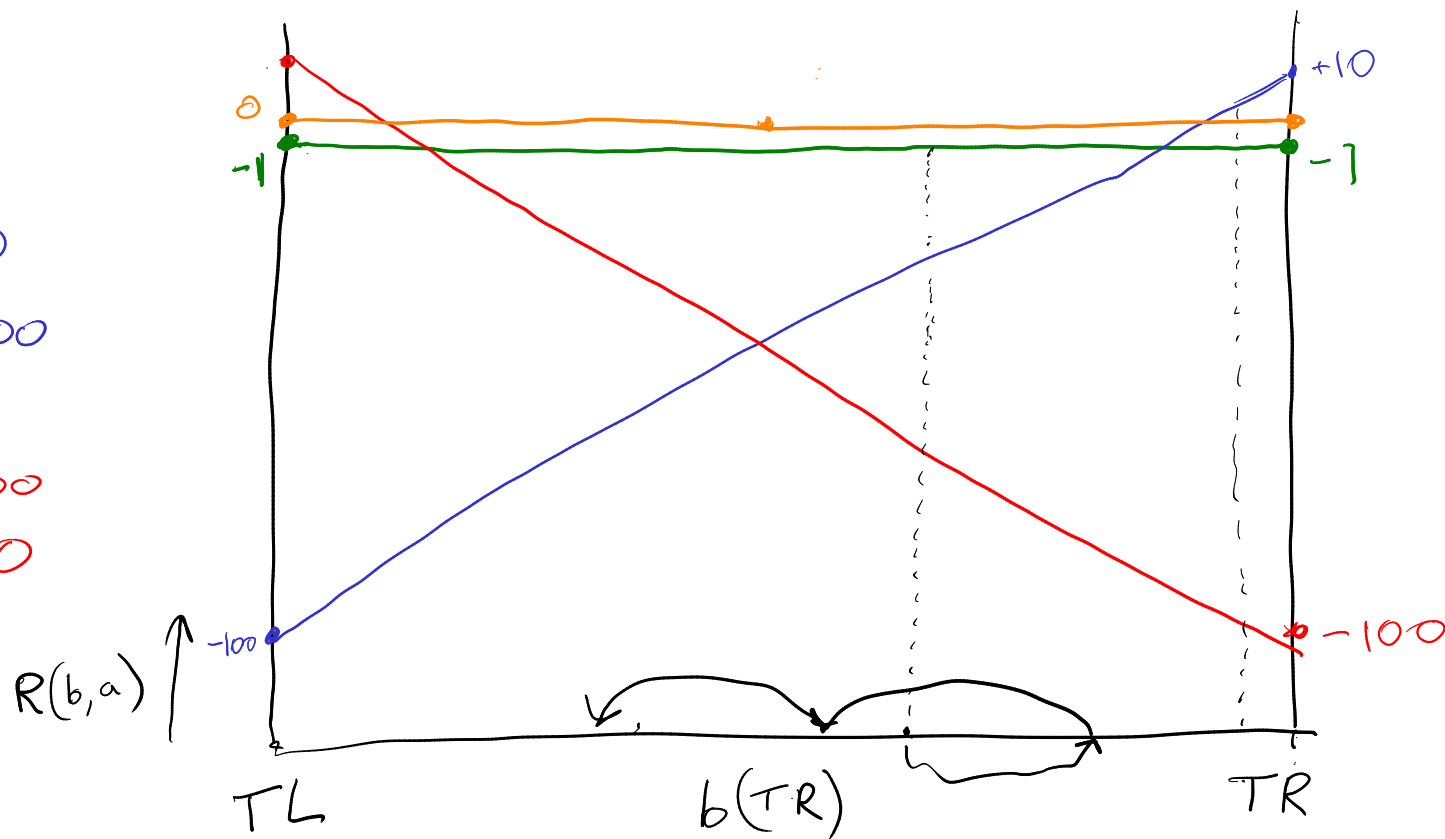
one-step
✓ α -vector
for action α

$$R(b, a) = \vec{R}^a \cdot \vec{b}$$

$\downarrow$
 $\begin{bmatrix} R(s^1, a) \\ R(s^2, a) \\ \vdots \\ R(s^n, a) \end{bmatrix}$

$\downarrow$
 $\begin{bmatrix} b(s^1) \\ b(s^2) \\ \vdots \\ b(s^n) \end{bmatrix}$

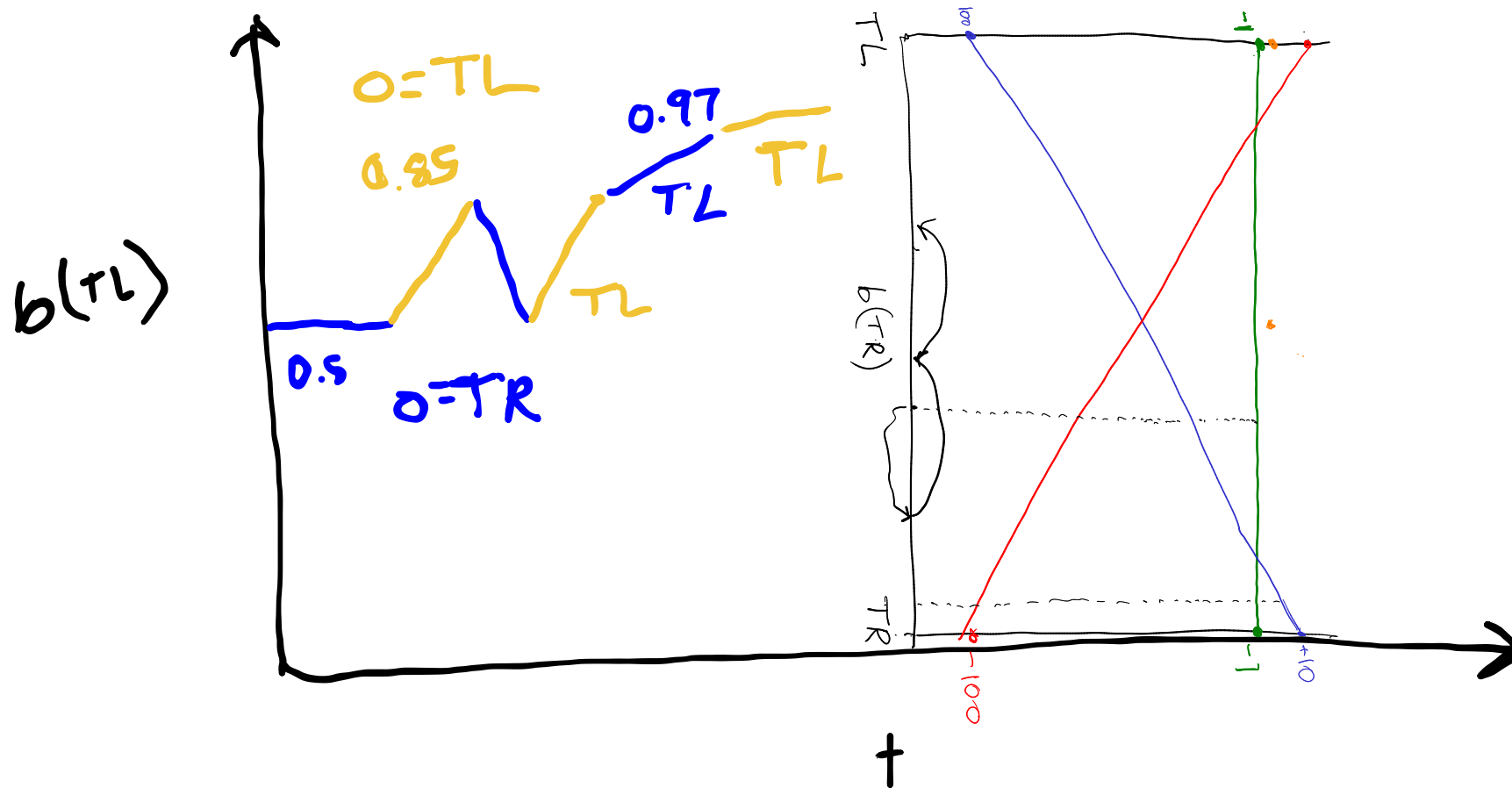
$$\pi_{\text{greedy}}(b) = \arg\max_a \vec{R}^a \cdot \vec{b}$$



$$R(b, a) = E[R(s, a)] = \sum_s b(s) R(s, a) = b(TL) R(TL, a) + (1 - b(TL)) R(TR, a)$$

α -vectors define a policy

$$b'(s') \propto Z(o \mid a, s') \sum_s T(s' \mid s, a) b(s)$$



Exercise 2: Crying Baby 1-Step Utility

$$S = \{h, \neg h\} \quad T(h \mid h, \neg f) = 1.0$$

$$A = \{f, \neg f\} \quad T(h \mid \neg h, \neg f) = 0.1$$

$$O = \{c, \neg c\} \quad T(\neg h \mid \cdot, f) = 1.0$$

$$R(s, a) = R(s) + R(a)$$

$$R(s) = \begin{cases} -10 & \text{if } s = h \\ 0 & \text{otherwise} \end{cases}$$

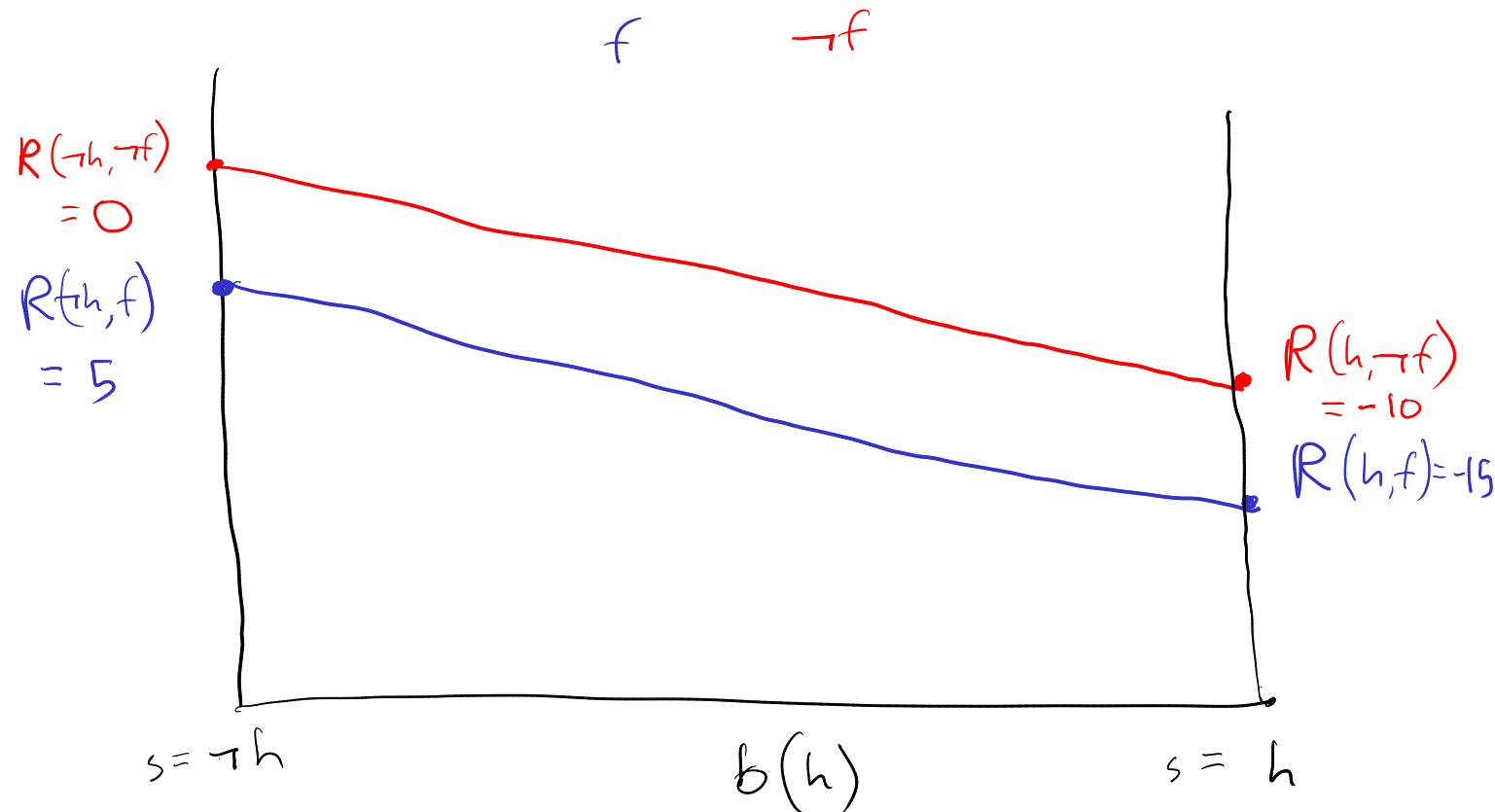
$$R(a) = \begin{cases} -5 & \text{if } a = f \\ 0 & \text{otherwise} \end{cases}$$

$$Z(c \mid \cdot, h) = 0.8$$

$$Z(c \mid \cdot, \neg h) = 0.1$$

$$\gamma = 0.9$$

Draw the 1-step utility α -vectors for the Crying Baby problem.



Alpha Vectors for Conditional Plans

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Conditional Plans: fixed-depth history-based policies

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1 Step:

Alpha Vectors for Conditional Plans

Conditional Plans: fixed-depth history-based policies

1 Step: (L) (oL) (oR)

Alpha Vectors for Conditional Plans

Conditional Plans: fixed-depth history-based policies

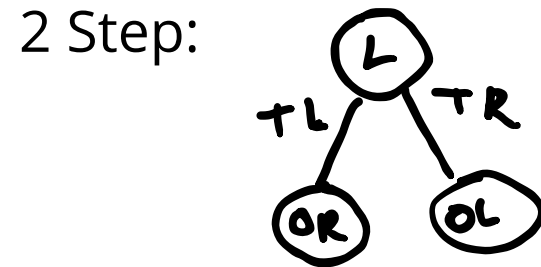
1 Step: (L) (oL) (oR)

2 Step:

Alpha Vectors for Conditional Plans

Conditional Plans: fixed-depth history-based policies

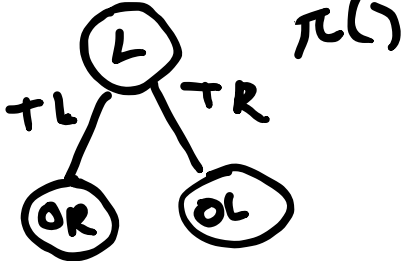
1 Step: (L) (OL) (OR)



Alpha Vectors for Conditional Plans

Conditional Plans: fixed-depth history-based policies

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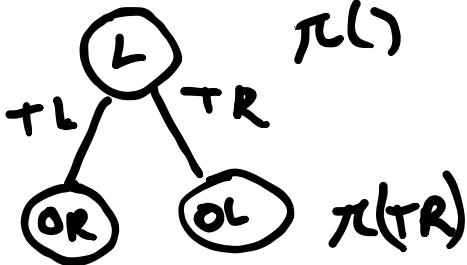
2 Step: 

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Conditional Plans: fixed-depth history-based policies

1 Step: (L) (OL) (OR)

2 Step:

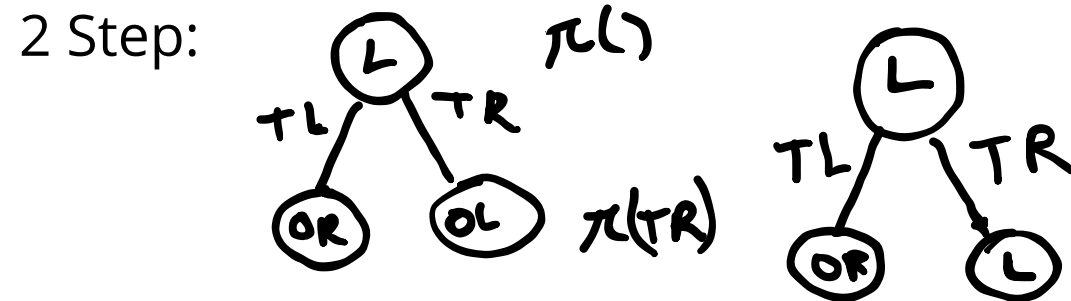


```
graph TD; L((L)) -- TL --> OR1((OR)); L -- TR --> OL((OL));
```

Alpha Vectors for Conditional Plans

Conditional Plans: fixed-depth history-based policies

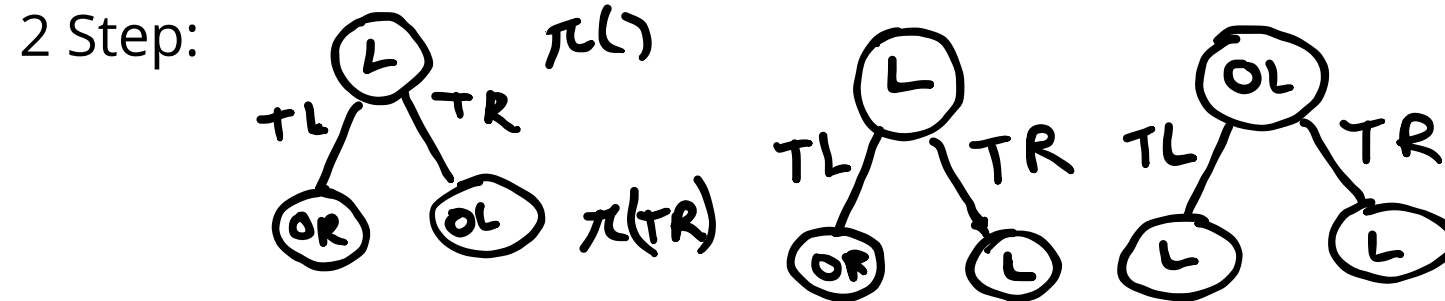
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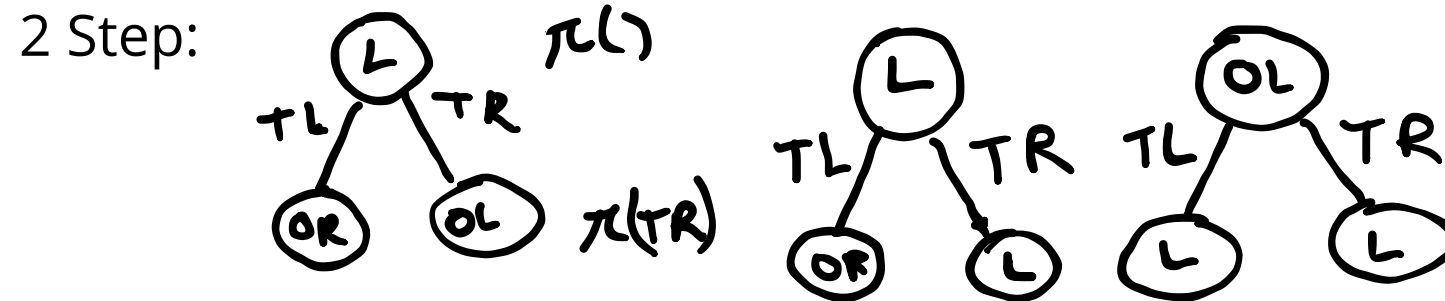
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Alpha Vectors for Conditional Plans

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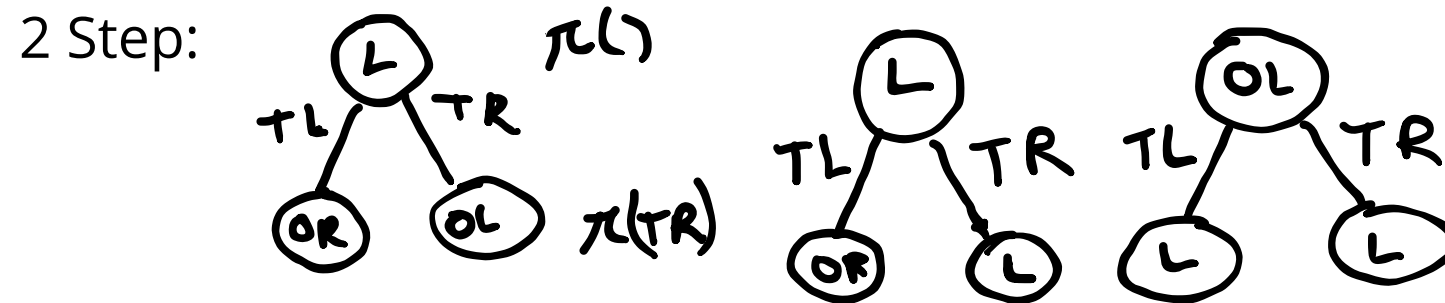


$$|A|^{\frac{(|O|^h - 1)}{(|O| - 1)}}$$

Alpha Vectors for Conditional Plans

Conditional Plans: fixed-depth history-based policies

1 Step: (L) (OL) (OR)



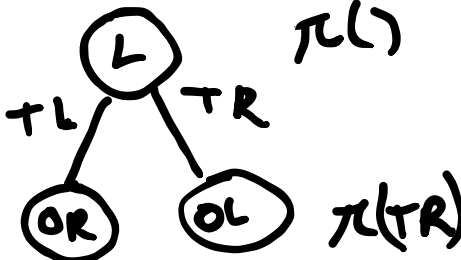
$$|A| \frac{(|O|^h - 1)}{(|O| - 1)}$$

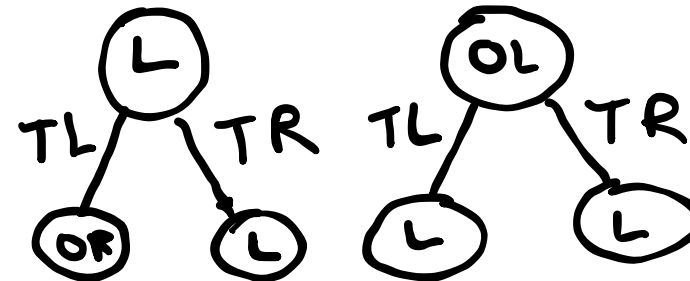
27 two step plans!

Alpha Vectors for Conditional Plans

Conditional Plans: fixed-depth history-based policies

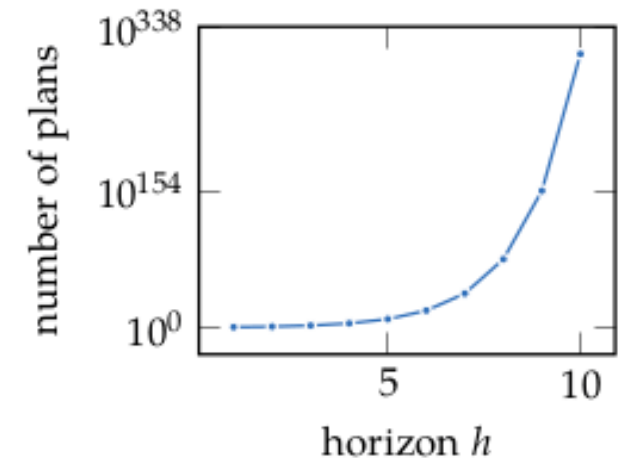
1 Step: (L) (OL) (OR)

2 Step:  $\pi(L)$ $\pi(TR)$



$$|A|^{\frac{(|O|^h - 1)}{(|O| - 1)}}$$

27 two step plans!

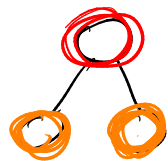


Alpha Vectors for Conditional Plans

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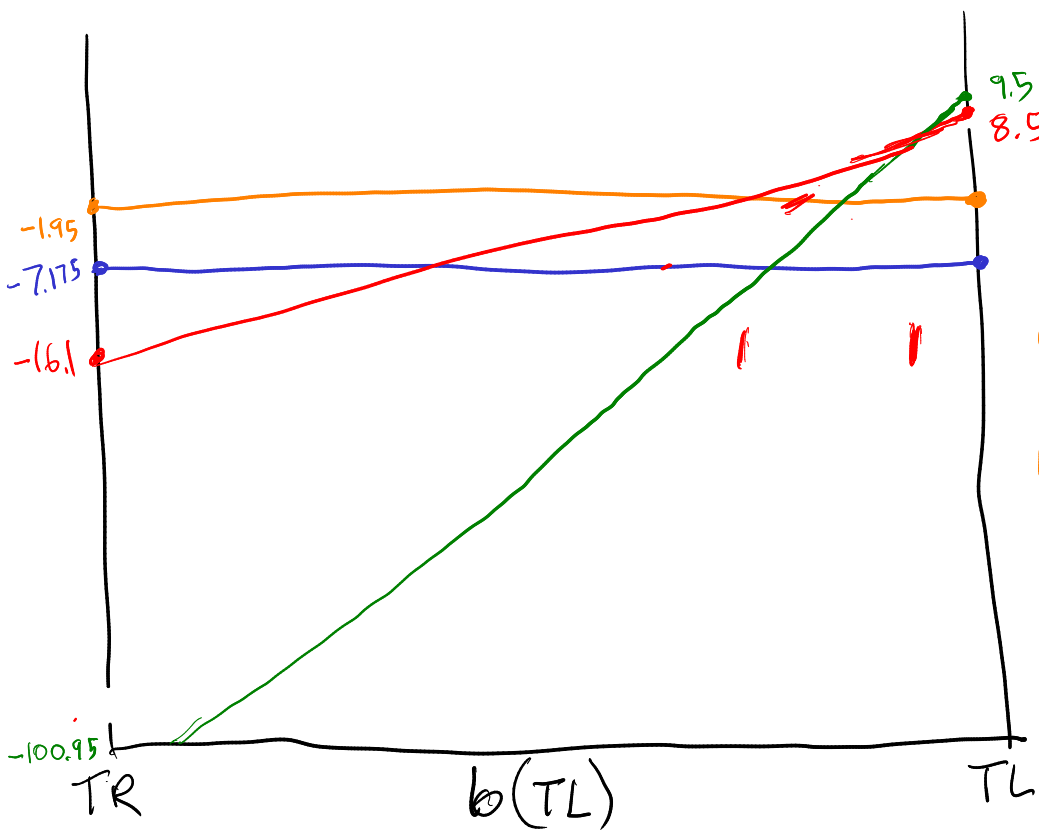
For 1-step: $U^\pi(s) = R(s, \pi())$

Alpha Vectors for Conditional Plans



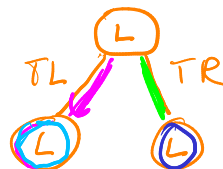
For 1-step: $U^\pi(s) = R(s, \pi())$

$$U^\pi(s) = \underline{R(s, \pi())} + \gamma \left[\sum_{s'} T(s' | s, \pi()) \sum_o \cancel{\pi(o | \pi(), s')} \underline{U^{\pi(o)}(s')} \right]$$

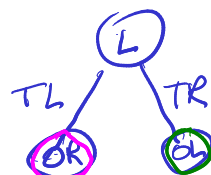


Ⓛ
 $U^{\text{Ⓛ}}(\cdot) = -1$

Ⓛ
 $U^{\text{Ⓛ}}(\text{TR}) = 10$
 $U^{\text{Ⓛ}}(\text{TL}) = -100$

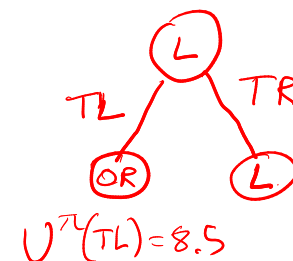


$U^\pi(\text{TL}) = -1 + \gamma [1(0.85(-1) + 0.15(-1))]$
 $= -1 + 0.95(-1) = -1.95$
 $U^\pi(\text{TR}) = -1.95$

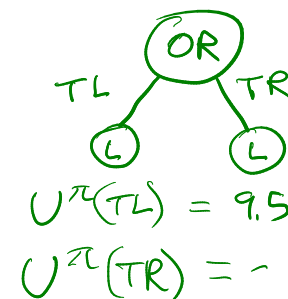


$U^\pi(\text{TL}) = -1 + \gamma [1(0.85 \cdot 10 + 0.15(-100))]$
 $= -7.175$
 $U^\pi(\text{TR}) = -7.175$

Ⓞ
 $U^{\text{Ⓞ}}(\text{TR}) = -100$
 $U^{\text{Ⓞ}}(\text{TL}) = 10$



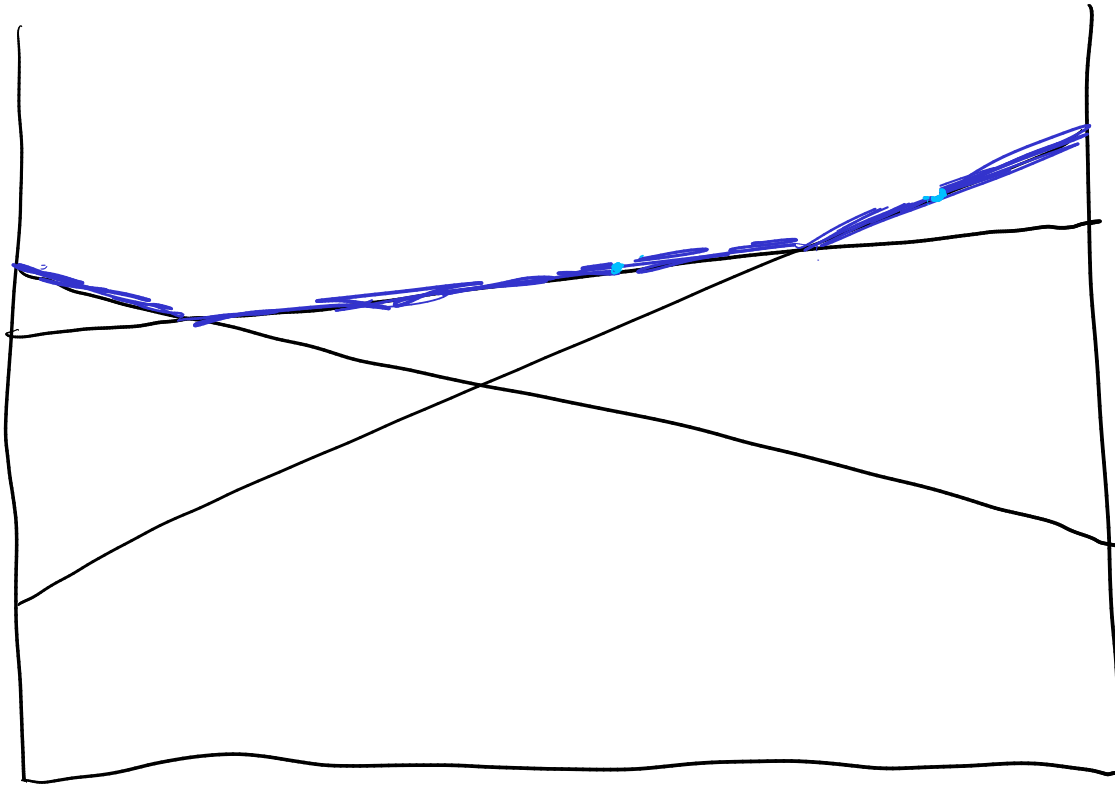
$U^\pi(\text{TL}) = 8.5$
 $U^\pi(\text{TR}) = -16$
 $-1 + \gamma(0.15(100) + 0.85(-1)) = -16.1$



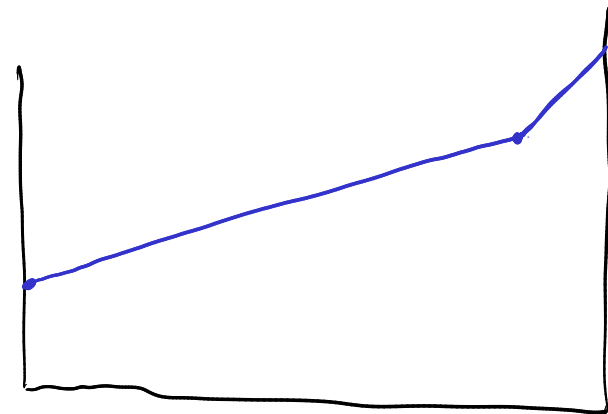
$U^\pi(\text{TL}) = 9.5$
 $U^\pi(\text{TR}) = -100.95$

POMDP Value Functions

POMDP Value Functions



$$\underline{V^*(b)} = \max_{\alpha \in \Gamma} \alpha^\top b$$



Exercise: 2-Step Crying Baby α Vectors

$$S = \{h, \neg h\}$$

$$T(h \mid h, \neg f) = 1.0$$

$$A = \{f, \neg f\}$$

$$T(h \mid \neg h, \neg f) = 0.1$$

$$O = \{c, \neg c\}$$

$$T(\neg h \mid \cdot, f) = 1.0$$

$$R(s, a) = R(s) + R(a)$$

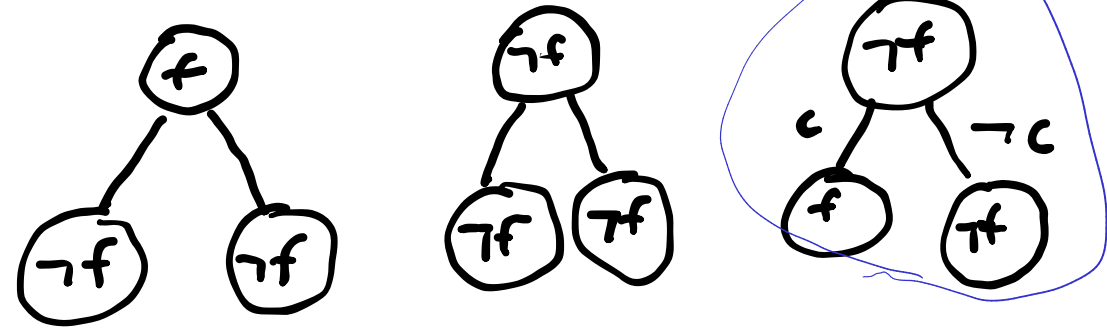
$$R(s) = \begin{cases} -10 & \text{if } s = h \\ 0 & \text{otherwise} \end{cases}$$

$$R(a) = \begin{cases} -5 & \text{if } a = f \\ 0 & \text{otherwise} \end{cases}$$

$$Z(c \mid \cdot, h) = 0.8$$

$$Z(c \mid \cdot, \neg h) = 0.1$$

$$\gamma = 0.9$$



$$U^\pi(s) = R(s, \pi()) + \gamma \left[\sum_{s'} T(s' \mid s, \pi()) \sum_o O(o \mid \pi(), s') U^{\pi(o)}(s') \right]$$

$$\textcircled{f} \\ U^f(h) = -15 \\ U^f(\neg h) = -5$$

$$\textcircled{\neg f} \\ U^{\neg f}(h) = -10 \\ U^{\neg f}(\neg h) = 0$$

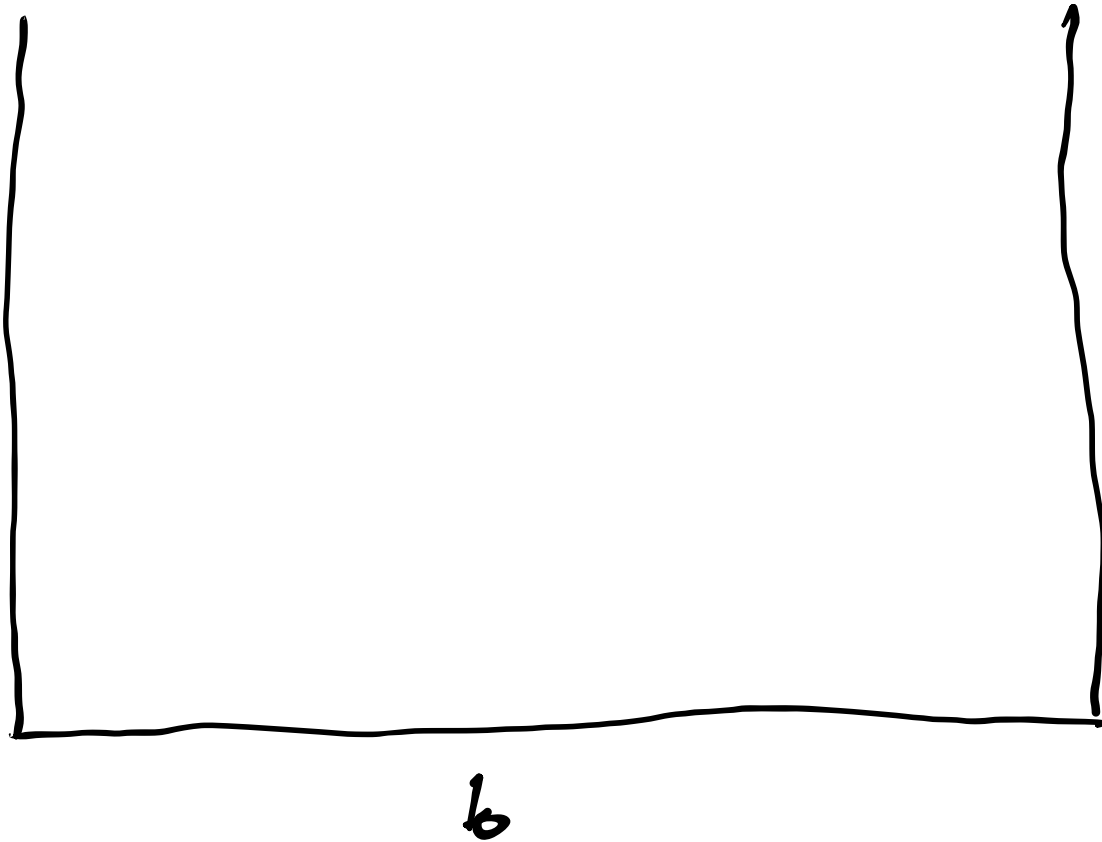
$$U^{\pi_A}(h) = R(h, \neg f) + \gamma \left[T(h \mid h, \neg f) \left(Z(c \mid \neg f, h) U^f(h) + Z(\neg c \mid \neg f, h) U^{\neg f}(h) \right) + T(\neg h \mid h, \neg f) \left(Z(c \mid \neg f, \neg h) U^f(\neg h) + Z(\neg c \mid \neg f, \neg h) U^{\neg f}(\neg h) \right) \right]$$

$$\begin{aligned} &= -10 + 0.9 [1(0.8(-15) + 0.2(-10))] = -22.6 \\ U^{\pi_A}(\neg h) &= R(\neg h, \neg f) + \gamma [T(h \mid \neg h, \neg f) (Z(c \mid \neg f, h) U^f(h) + Z(\neg c \mid \neg f, h) U^{\neg f}(h)) + T(\neg h \mid \neg h, \neg f) (Z(c \mid \neg f, \neg h) U^f(\neg h) + Z(\neg c \mid \neg f, \neg h) U^{\neg f}(\neg h))] \\ &= 0 + 0.9 [0.1(0.8(-15) + 0.2(-10)) + 0.9(0.1(-5) + 0.9(0))] = -1.665 \end{aligned}$$

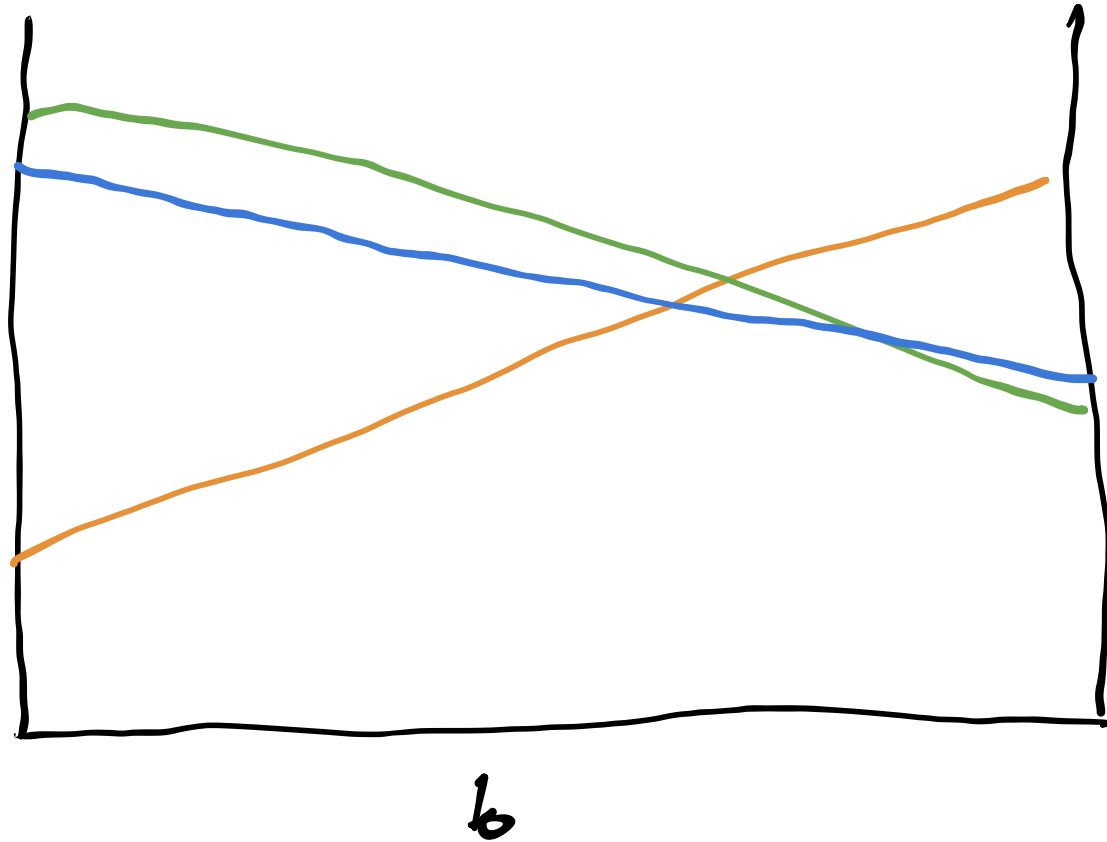
$$\alpha_{\pi_A} = [-22.6, -1.665]$$

α -Vector Pruning

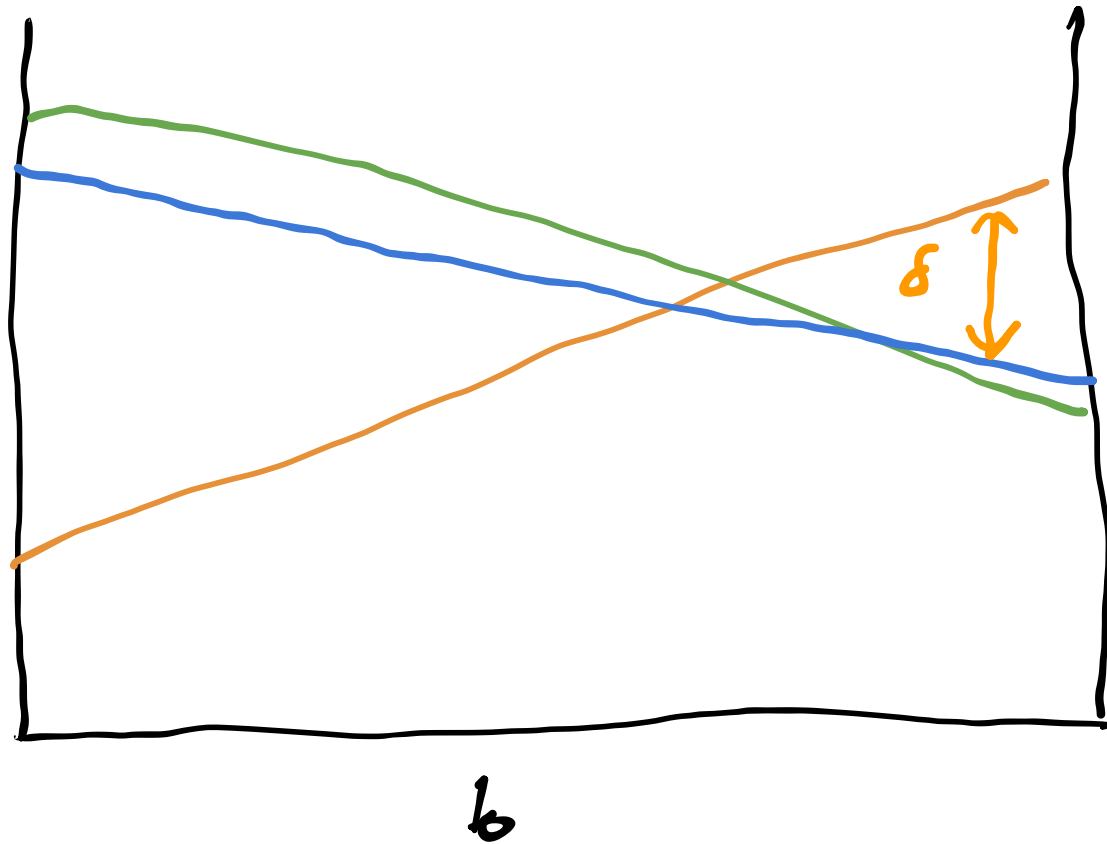
α -Vector Pruning



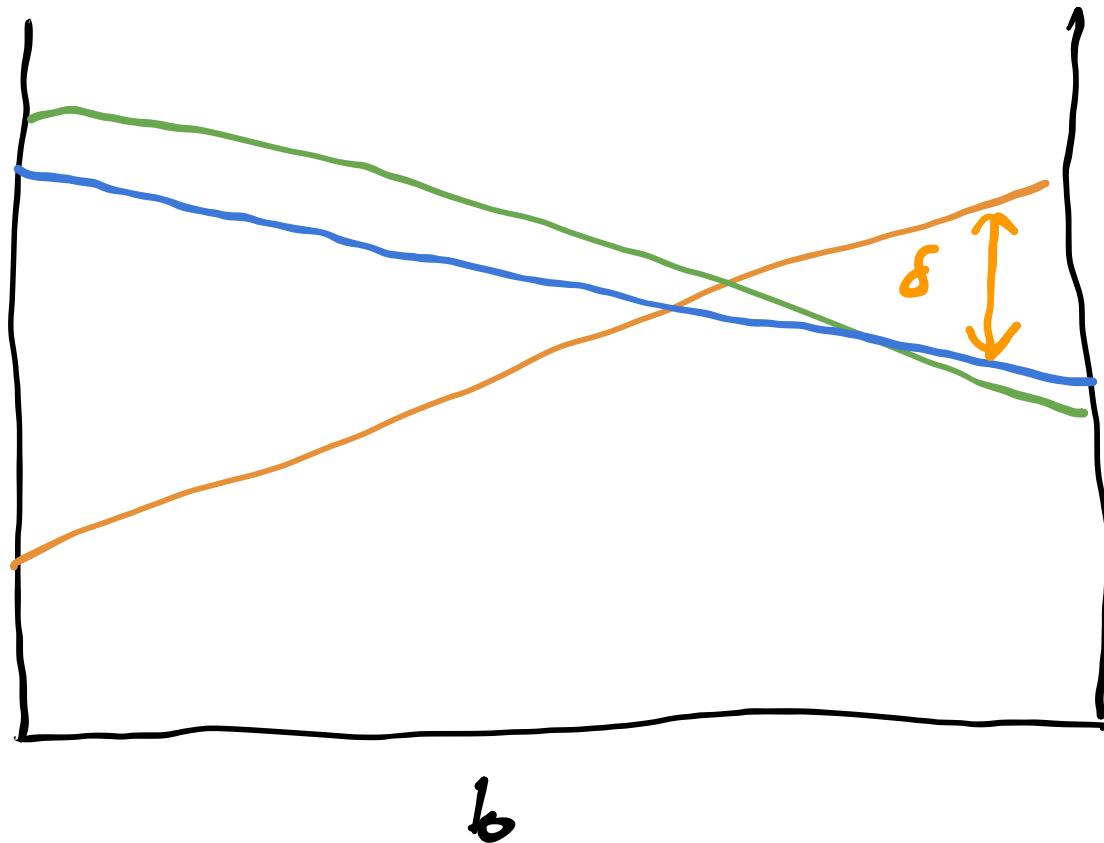
α -Vector Pruning



α -Vector Pruning

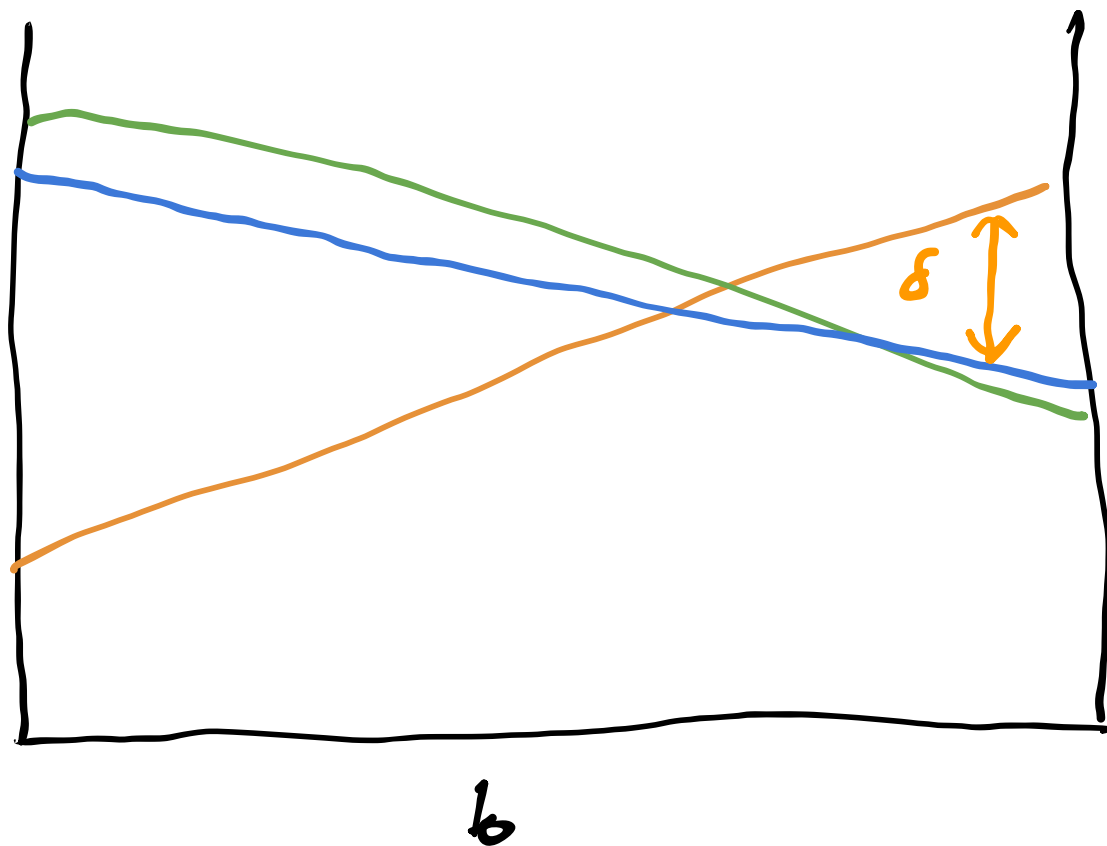


α -Vector Pruning



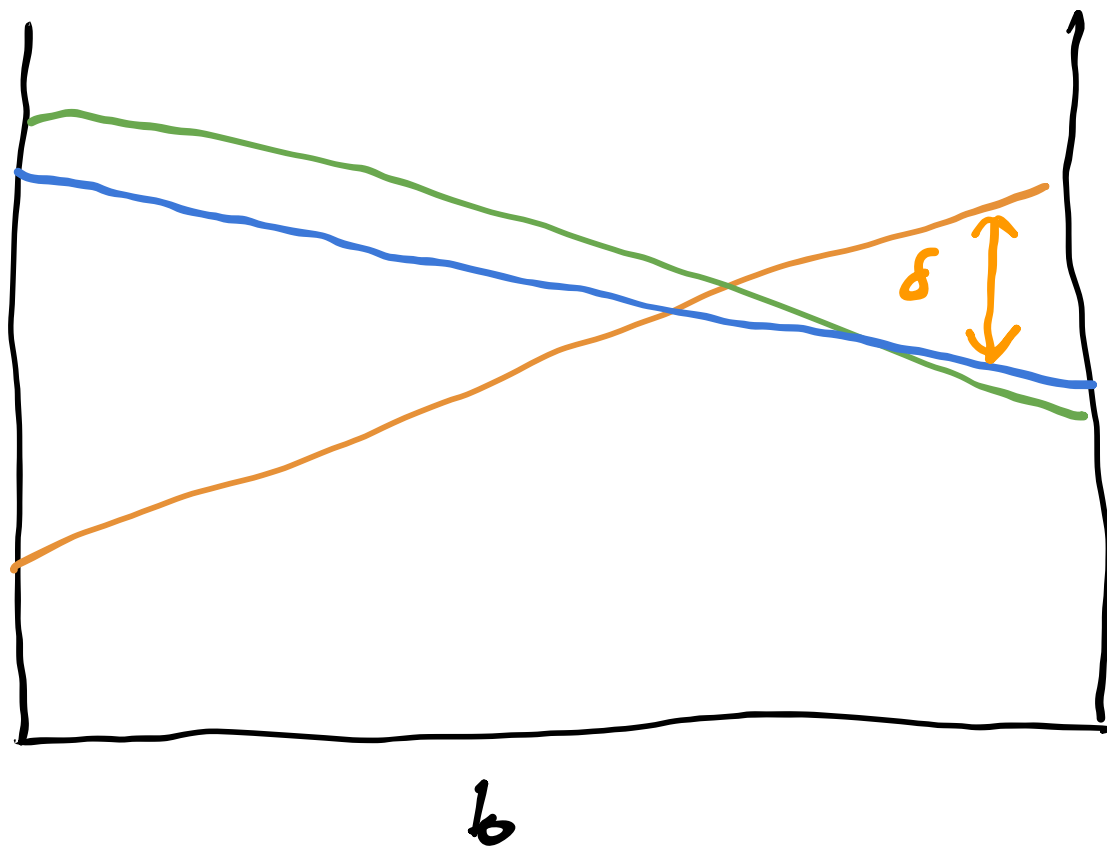
maximize δ
 δ, b

α -Vector Pruning



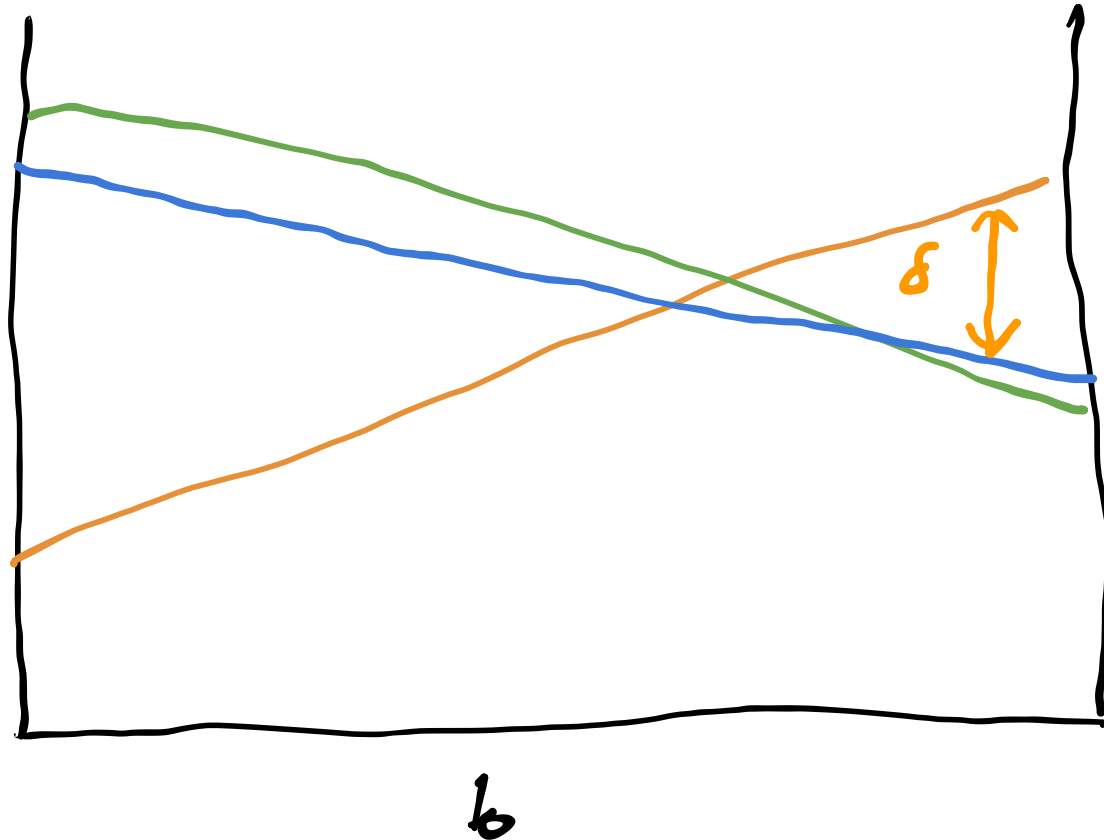
maximize δ
subject to $b \geq 0$

α -Vector Pruning



$$\begin{aligned} & \underset{\delta, b}{\text{maximize}} && \delta \\ & \text{subject to} && b \geq 0 \\ & && \mathbf{1}^\top b = 1 \end{aligned}$$

α -Vector Pruning

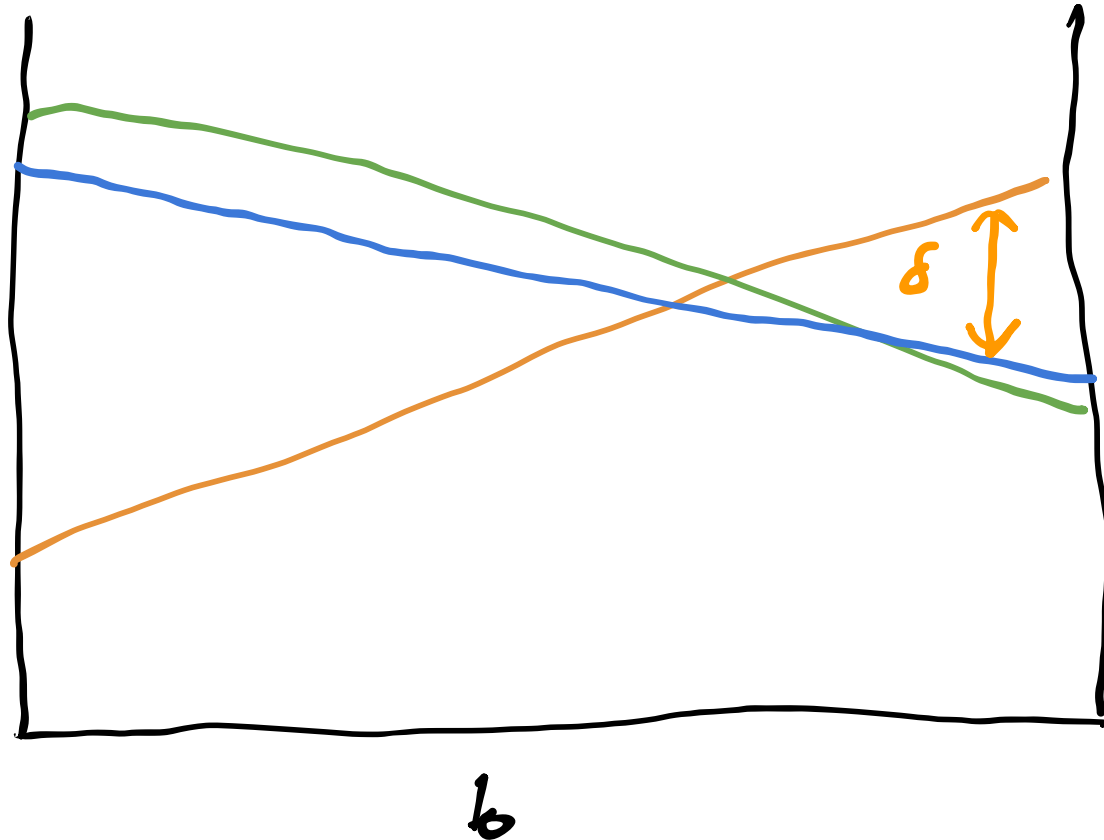


$$\begin{aligned} &\underset{\delta, b}{\text{maximize}} && \delta \\ &\text{subject to} && b \geq 0 \end{aligned}$$

$$\mathbf{1}^\top b = 1$$

$$\alpha^\top b \geq \underline{\alpha'}^\top b + \delta \quad \forall \alpha' \in \underline{\Gamma}$$

α -Vector Pruning



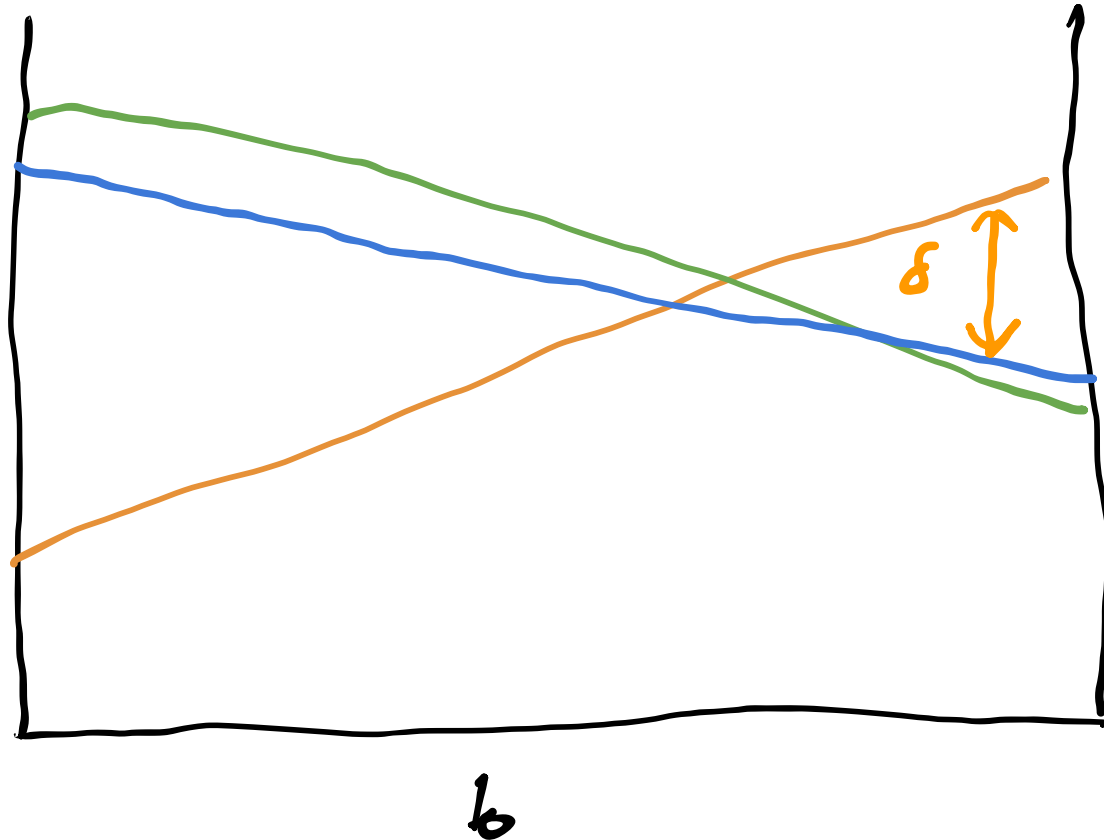
maximize δ
subject to $b \geq 0$

$$\mathbf{1}^\top b = 1$$

$$\alpha^\top b \geq \alpha'^\top b + \delta \quad \forall \alpha' \in \Gamma$$

"Linear Program": easy
optimization problem

α -Vector Pruning



maximize δ
subject to $b \geq 0$

$$\mathbf{1}^\top b = 1$$

$$\alpha^\top b \geq \alpha'^\top b + \delta \quad \forall \alpha' \in \Gamma$$

"Linear Program": easy
optimization problem

If $\delta^* > 0$, α is not dominated; b
solution sometimes called
"witness".

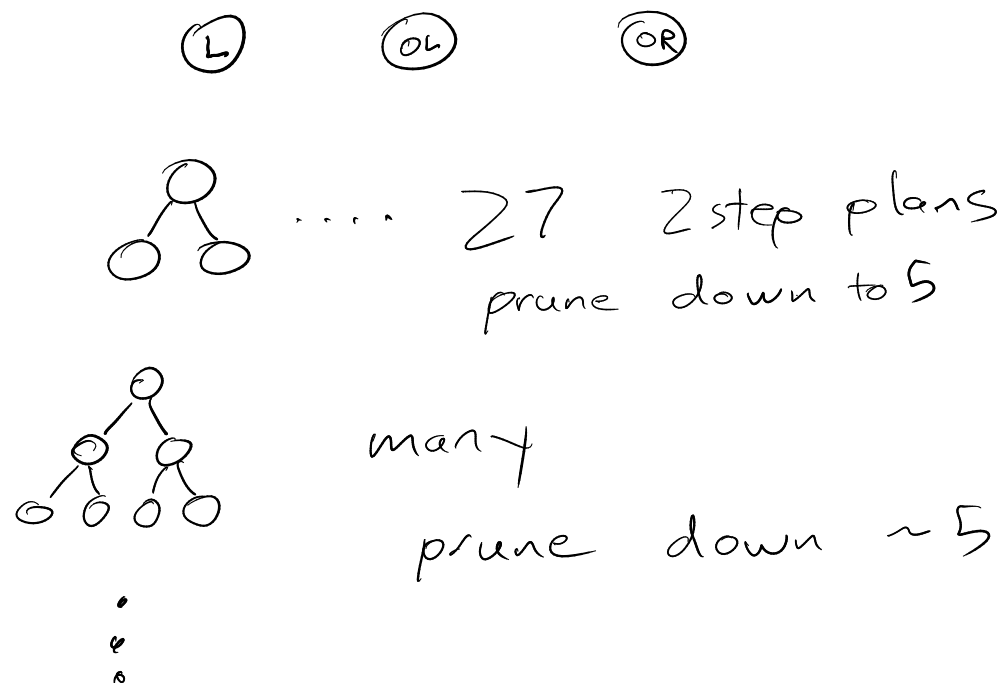
POMDP Value Iteration (horizon d)

$\Gamma^0 \leftarrow \emptyset$

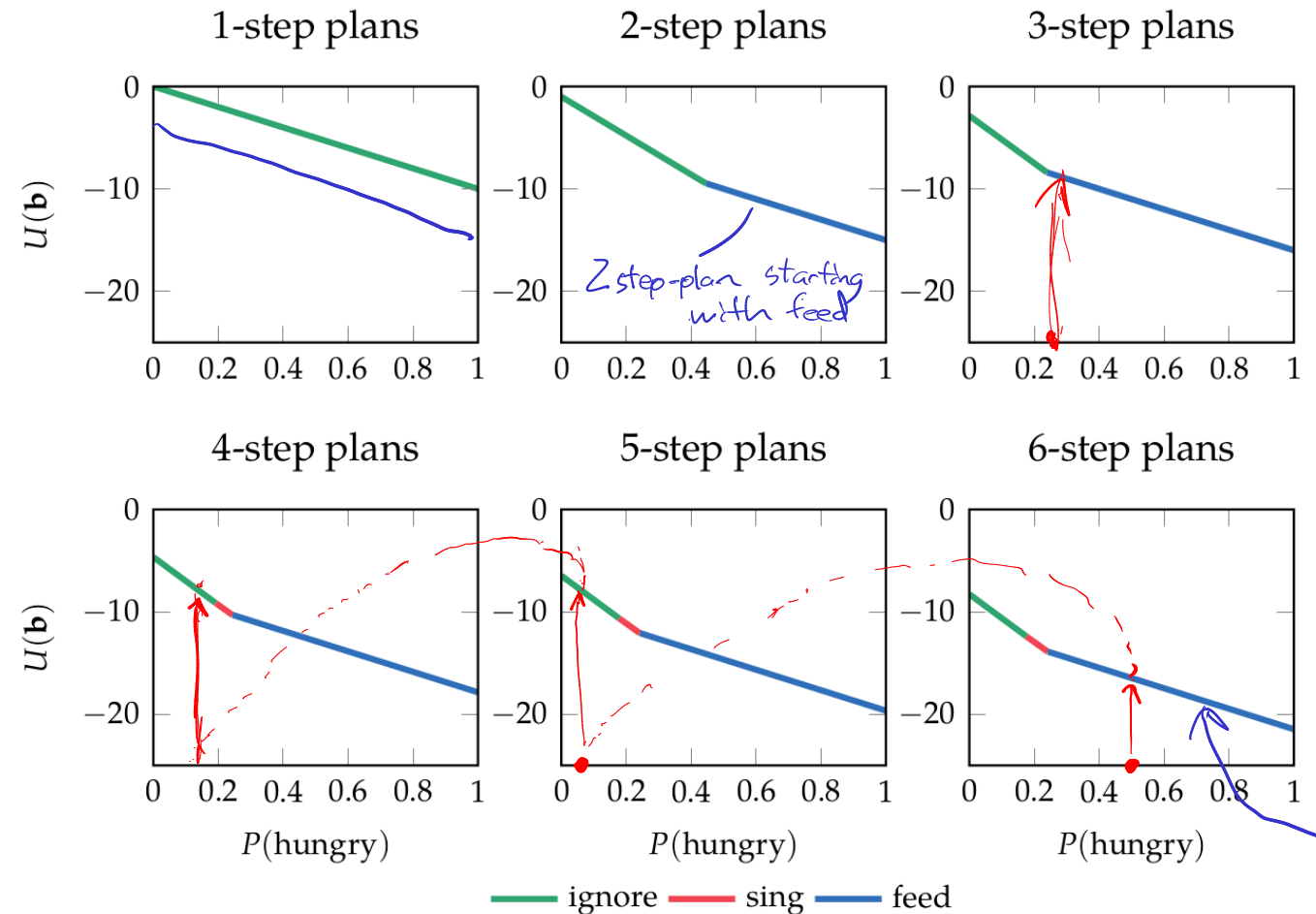
for $n \in 1 \dots d$

Construct Γ^n by expanding with Γ^{n-1}

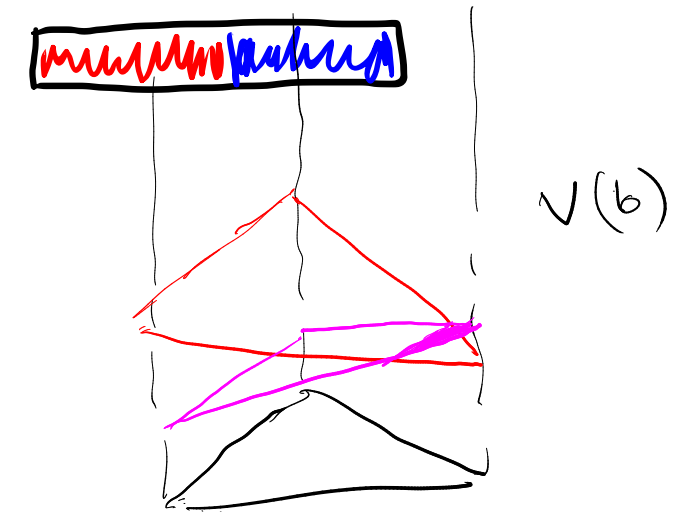
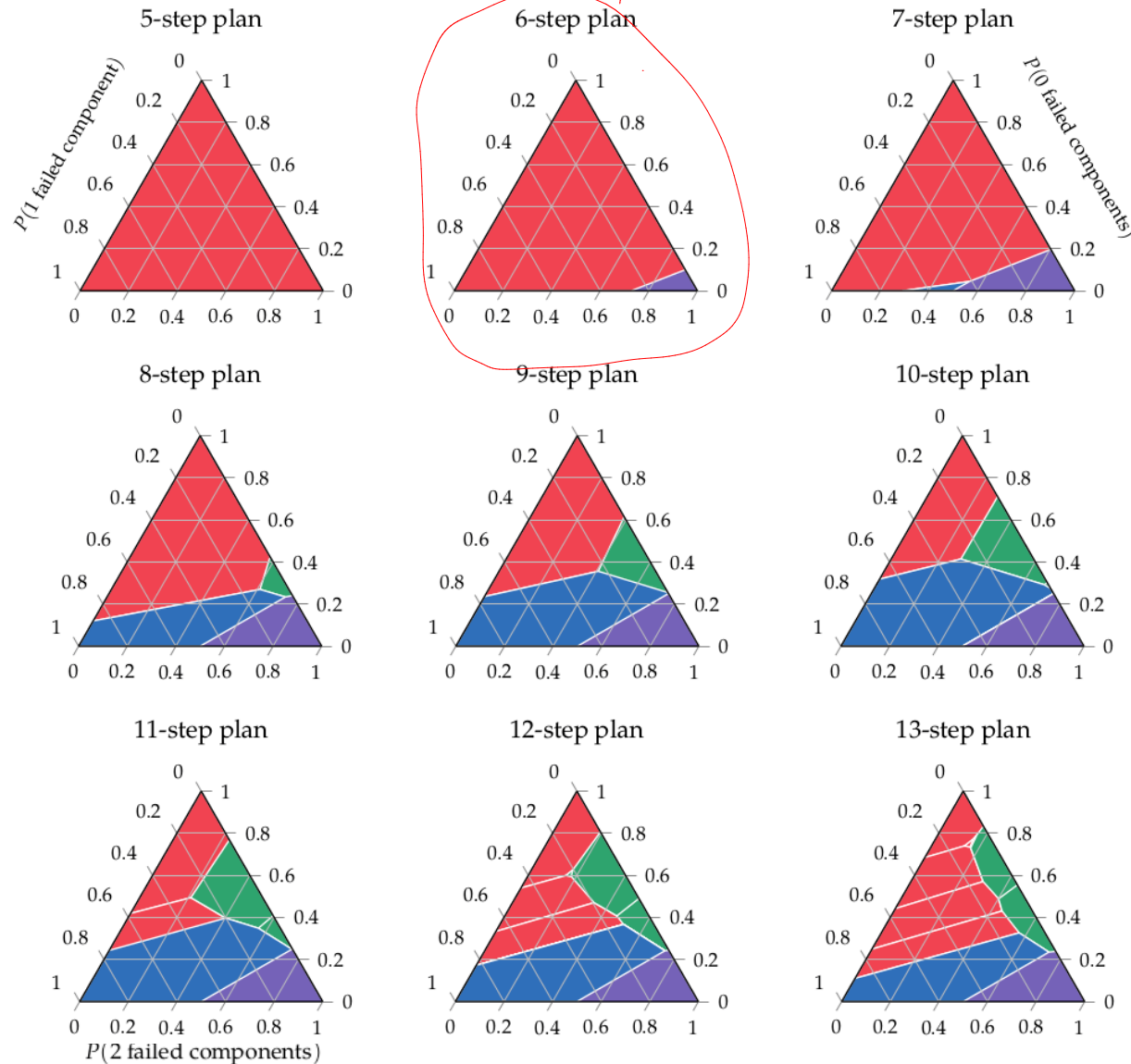
Prune Γ^n



Finite Horizon POMDP Value Iteration



Finite Horizon POMDP Value Iteration



Recap

Recap

- A POMDP is an MDP on the _____

Recap

- A POMDP is an MDP on the belief space

Recap

- A POMDP is an MDP on the belief space
- The value function of a discrete POMDP can be represented by a set of _____

Recap

- A POMDP is an MDP on the belief space
- The value function of a discrete POMDP can be represented by a set of α -vectors

Recap

- A POMDP is an MDP on the belief space
- The value function of a discrete POMDP can be represented by a set of α -vectors
- Each α vector corresponds to a _____

Recap

- A POMDP is an MDP on the belief space
- The value function of a discrete POMDP can be represented by a set of α -vectors
- Each α vector corresponds to a conditional plan